Revision Table

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2	Added LaTeX packages for code and formatting	Preamble
3	Restructured to match approved ClassiFish format	Chapter organization
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Table 1: Thesis Revision History

Consol: A Notes-to-Recollection Application Using SimCSE for its Textual Semantic Comparison

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Abstract

Generative AI has presented society with an opportunity to create a surplus of educational content in mere seconds. While knowledge is inherently beneficial and promotes intellectual and academic growth, there still is a bridge to actual learning that cannot be crossed just through using AI and its instantaneous generative abilities (Mishra & Anthony, 2023). Learning, in the academic landscape, is comprised of obtaining information which can be in the form of notes (Rashid & Rigas, 2006; Devlin et al., 2019). Retaining knowledge after note-taking remains a challenge, as forgetting is inevitable and overlooked by recent AI and NLP advancements. (Klemm, 2012; Naim et al., 2020). In studying and reviewing, if a person wishes to ensure their recollection of key topics is accurate and coherent, options are limited without manually referencing source material, which can be tiring and inefficient. Assistance is needed, particularly in providing instant feedback for accuracy and mastery of notes (Karpicke & Roediger, 2008; Sachs, 1967). This research proposes Consol, a notes-to-recollection application that uses SimCSE's contrastive sentence embeddings and semantic similarity analysis to compare how similar a person's recollection is to their original notes (Gao et al., 2021; Corley et al., 2005), and uses progress tracking techniques to encourage consistency and improvement (Deterding et al., 2011; Gaston & Cooper, 2017). It will be a web-based application that leverages semantic similarity evaluation to assess memory recall effectively. The study aims to evaluate whether semantic-based feedback enhances learning outcomes compared to traditional recall methods. Through system testing and user performance assessment, this research will analyze the system's usability, accuracy, and effectiveness in improving memory retention (Baars et al., 2022; Tran & Nguyen, 2022). The results aim to provide a scalable solution bridging note-taking and active learning while fostering consistent knowledge consolidation (Vișcu, 2024; Stojanovic et al., 2020).

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1 Introduction

Memory is not merely the result of recording information but the outcome of consistent review and active engagement with what has been learned. Notes, while essential for capturing ideas, do not inherently lead to long-term retention or understanding simply by being created. Instead, it is through revisiting, processing, and consolidating these notes that individuals truly reinforce their knowledge (Karpicke and Roediger, 2008; Klemm, 2012).

For students, teachers, and professionals alike, the challenge lies in establishing consistent learning practices and receiving validation for their efforts; validation that not only measures recall but also provides meaningful feedback on comprehension (Stojanovic et al., 2020; Baars et al., 2022). Traditional note-taking methods often focus on transcription, prioritizing the exact wording of information over its deeper meaning. However, research by Sachs (1967) highlights that individuals naturally retain the semantic meaning; the "gist" of information; rather than its verbatim form.

This tendency to paraphrase and restructure ideas reflects the cognitive process of abstraction, where understanding takes precedence over memorization. Conventional systems that evaluate recall by word-for-word accuracy fail to account for this natural process, offering an incomplete and often discouraging measure of learning performance (Corley et al., 2005; Gao et al., 2021).

Additionally, progress tracking has proven to be an effective strategy for sustaining motivation and fostering consistent learning habits. By incorporating elements such as performance metrics, mastery indicators, and progress tracking, such systems provide users with actionable insights and a sense of achievement (Deterding et al., 2011; Gaston and Cooper, 2017). These tools not only validate the user's learning progress but also make the process more interactive and rewarding, promoting deeper engagement and long-term retention (Tran and Nguyen, 2022; Viscu, 2024).

1.1 | Background of the Study

Now, as technology continues to advance in the current "information age," the demand for fast, accurate, and easily accessible information has increased dramatically. This is particularly evident in fields where knowledge acquisition is critical, such as education, research, and personal development. The proliferation of generative AI technologies and digital tools has revolutionized the ways individuals gather, process, and manage information. Digital note-taking tools, such as Evernote, Notion, Obsidian, and various AI-assisted applications, offer features like automatic transcription, cloud synchronization, and summarization, making the recording of ideas and knowledge more efficient and convenient to the modern user (Baars et al., 2022; Rashid and Rigas, 2006).

Existing studies confirm that effective learning requires more than passive note creation. Research conducted by Sachs (1967) indicates that individuals are better at retaining semantic meaning rather than exact syntactic details. This means that during memory retrieval, individuals often paraphrase, restructure, and abstract ideas rather than reproduce content verbatim. Traditional systems that rely on word-for-word accuracy to evaluate recall fail to account for this natural cognitive process, leading to an incomplete understanding of memory performance (Corley et al., 2005; Gao et al., 2021).

Building on this, the integration of AI-driven systems in learning environments has opened new avenues for assessing memory and comprehension more effectively. By including the more recent advancements in natural language processing (NLP) and similarity-based algorithms, these systems can evaluate recall beyond mere verbatim reproduction, focusing instead on the semantic alignment between original and recalled content (Devlin et al., 2019; Gao et al., 2021). Such approaches recognize the inherent flexibility of human cognition, where meaningful understanding takes precedence over exact replication. For example, sentence-transformer models like SimCSE have demonstrated the ability to measure semantic similarity with high accuracy, making them suitable for applications that assess memory recall in a way that mirrors human evaluation (Gao et al., 2021; Corley et al., 2005).

This shift toward semantic-based assessment not only provides a more accurate measure of knowledge retention but also aligns with modern pedagogical principles that emphasize deep learning and critical thinking. By rewarding comprehension and conceptual mastery over rote memorization, AI-assisted systems encourage learners to engage more actively with the material, fostering long-term knowledge retention and improved cognitive performance (Karpicke and Roediger, 2008; Klemm, 2012). As these technologies continue to evolve, they hold the potential to transform traditional methods of learning evaluation and pave the way for more adaptive and personalized educational tools (Stojanovic et al., 2020; Viscu, 2024).

1.2 | Statement of the Problem

The rapid development and advancement of Generative AI has led to the overwhelming abundance of knowledge, and with that, an unlimited opportunity to take notes. While this is clearly for some benefit, it is not put into actual use if the increasing number of notes are not consolidated into memory, thereby hindering the cumulative learning process (Karpicke & Roediger, 2008; Klemm, 2012). Additionally, in individual study contexts, there is a lack of mechanisms to verify whether a learner has truly understood or mastered the material. This problem is compounded by the fact that human recall often involves paraphrasing and abstracting information, making it difficult for current systems to accurately assess a learner's understanding (Sachs, 1967; Corley et al., 2005). Consequently, there is a need for a system that can evaluate learning by recognizing and comparing both the original content and its reconstructed ideas through context clues, ensuring that abstracted yet relevant concepts are acknowledged without overlooking their presence in the material (Gao et al., 2021; Devlin et al., 2019).

1.3 | Objectives

This section outlines the primary goals of the study, leaning more into creating a system that acts as a proof-of-concept for the use of semantic comparison within the context of evaluating a user's degree of recollection.

1.3.1 | General Objective

To develop a web-based memorization system that refers to the user's pre-made notes and prompts the user to perform active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas.

1.3.2 | Specific Objectives

1. Design the system architecture of the notes-to-recollection comparison system, including its scoring criteria for evaluating recall accuracy.
2. Devise pre-processing procedures for removing unnecessary non-alphanumeric symbols and formatting artifacts to be suitable for semantic comparison.
3. Employ Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts.
4. Implement a web-based application integrating the SimCSE Model, giving it an appropriate user experience that facilitates motivation through progress tracking and active learning.
5. Conduct a usability test on the web-based application and document the results.

1.4 | Scope and Limitation

The study is strictly limited to a maximum of 10 test participants. The study will be conducted from January 2025 through December of the same year.

1.5 | Significance and Contribution

The proposed implementation will assist users in memorizing their notes and incentivize steady progress in content mastery through various tracking strategies.

This research contributes to the field through several key innovations: the first educational application of SimCSE contrastive learning for semantic similarity assessment, demonstrating novel integration of unsupervised contrastive learning principles with educational technology; development of a multi-dimensional assessment framework that evaluates recall through semantic similarity, completion metrics, and temporal performance indicators; implementation of memory consolidation theory through practical digital application, bridging cognitive science principles with educational technology; and advancement of objective educational assessment by reducing subjective bias through consistent semantic evaluation.

The system's name "Consol" reflects the fundamental educational principle of memory consolidation, where the brain naturally compresses and abstracts information for long-term storage. Rather than penalizing students for paraphrasing or conceptual restructuring, the system recognizes that consolidation inherently involves "compaction of bulk data," allowing semantic understanding to take precedence over verbatim reproduction. This approach aligns with natural cognitive processes while maintaining academic rigor through objective similarity measurement.

The research addresses critical gaps in self-study technology where traditional assessment methods somewhat fail to completely account for semantic equivalence, providing a foundation for future adaptive learning systems that recognize and reward genuine comprehension over superficial memorization.

2

Review of Related Literature

This chapter reviews relevant literature on note-taking strategies, memory processes, natural language processing (NLP) for semantic similarity, and progress tracking techniques. These areas provide the theoretical basis for understanding information encoding, memory consolidation, textual comparison, and user engagement. By examining these interconnected concepts, the review establishes a foundation for exploring methods that enhance learning, recall, and performance. The studies discussed offer insights into optimizing knowledge retention, evaluating recall accuracy, and motivating users through structured feedback mechanisms.

2.1 | Memory and Cognitive Processes

Memory plays a crucial role in the learning process, specifically in the retention and recall of information. Studies have demonstrated that memory prioritizes the retention of meaning over surface-level details, which has direct implications for comprehension and learning.

2.1.1 | Working Memory and Learning Efficiency

Klemm (2007) establishes working memory as the cognitive foundation for intellectual performance, functioning as a "scratch pad" where information chunks are processed during thinking operations. The tight correlations between working memory capacity, IQ, and problem-solving abilities demonstrate that students' learning effectiveness depends on their ability to maintain and manipulate information during cognitive tasks, challenging contemporary educational approaches that emphasize understanding while neglecting memorization skills.

This research reveals that robust memory expands cognitive capabilities and expedites new understanding development. Students cannot effectively apply concepts they understand if they cannot remember them when needed, establishing memory as essential for both knowledge retention and intellectual development rather than mere rote learning.

2.1.2 | The Role of Semantic and Syntactic Memory

Sachs (1967) explored the relationship between semantic and syntactic memory in the context of comprehension. In her study, participants were presented with sentences and tested on their ability to recognize changes in meaning and form after brief intervals. The findings revealed that memory retention is primarily focused on semantic content, while syntactic details, such as active-to-passive transformations, fade quickly. Even when participants retained the overall meaning of the sentences, the formal properties of the sentences were quickly forgotten. This study concludes that memory is structured to prioritize the "deep structure" or meaning of a sentence over its "surface structure" or syntax, suggesting that comprehension relies on extracting and retaining the core meaning of information.

Sachs' temporal dynamics research reveals that after 80-160 syllables of interpolated material, recognition for syntactic changes dropped to near chance levels while semantic recognition remained robust, demonstrating that individuals naturally retain meaning rather than exact form. This natural tendency toward semantic retention over syntactic precision provides theoretical justification for similarity-based evaluation systems that align with natural memory processes, supporting assessment approaches that recognize semantic equivalence while maintaining academic rigor.

2.1.3 | Cognitive Processes in Memory Retention

Rashid and Rigas (2006) provide an analysis of memory stages and their relationship to learning and information retention. The study outlines the progression of information through sensory memory, short-term memory, and long-term memory, highlighting the necessity of active cognitive processing for effective retention. Encoding, which involves the initial capture of relevant information, and reviewing, which reinforces the transfer of information to long-term memory, are identified as critical phases in the learning process. This study also discusses retrieval failure as a cause of forgetting, where proactive interference and retroactive interference disrupt recall.

Proactive interference occurs when previously learned information impairs the learning of new material, while retroactive interference arises when newly acquired knowledge interferes with the recall of older information. These findings emphasize that memory retention is optimized when learners engage in deep processing activities, such as organizing and summarizing information, rather than relying on surface-level strategies. This perspective aligns with the findings of Sachs (1967), reinforcing the importance of meaningful encoding and the prioritization of core content over superficial details.

2.1.4 | Memory and Recall

Naim et al. (2020) identified that recall performance grows sublinearly as the list length increases. This sublinear growth highlights a saturation effect, where the number of items retrieved plateaus despite an increase in encoded material. The study also revealed that encoding quality plays a crucial role in determining recall performance. Slower presentation rates allowed participants to encode a higher number of words into memory, which improved the total recall capacity. These findings provide evidence that memory recall adheres to universal laws, offering a structured and deterministic perspective on retrieval processes.

2.1.4.1 | Retrieval-Based Testing

Karpicke and Roediger (2008) emphasize that testing is not merely a means of assessment but a potent strategy for enhancing long-term memory retention. Their study debunks the conventional belief that once information is initially learned, further practice adds little value. Instead, repeated retrieval; through testing; promotes deeper encoding and strengthens memory consolidation, highlighting retrieval as an active mechanism essential for sustained learning performance.

2.1.4.2 | Optimizing Retrieval Practice for Long-term Retention

Karpicke and Roediger (2008) demonstrated that retrieval practice plays a central role in consolidating memories and improving long-term retention. Their experiments revealed that repeated retrieval significantly outperforms repeated study in enhancing memory performance. This finding validates the importance of active recall over passive encoding strategies.

The systematic investigation by Karpicke and Roediger reveals critical distinctions between study repetition and test repetition effects that challenge conventional educational assumptions. Once students correctly produce learning items, continued studying provides no measurable retention benefit, whereas continued testing generates substantial positive effects on long-term memory consolidation. This differential impact suggests that retrieval practice engages cognitive mechanisms distinct from those involved in passive review.

Furthermore, students' predictions of their own learning performance show no correlation with actual retention outcomes, indicating poor metacognitive awareness of effective learning strategies. This metacognitive blindness suggests that students may persist with ineffective study methods while avoiding retrieval practice that would substantially improve their learning outcomes. These findings provide empirical support for educational systems that prioritize active recall over passive content review.

2.1.5 | Mathematical Models of Memory Recall

Naim et al. (2020) demonstrate that human memory recall, despite appearing fragile and unpredictable, follows universal mathematical laws that can be derived from first principles. Their research on free recall paradigms reveals that memory operates as a deterministic process rather than a random phenomenon, with recall performance following predictable mathematical relationships.

The fundamental law of memory recall establishes that the average number of items recalled (R) from M items in memory follows the relationship $R = \sqrt{\frac{3\pi M}{2}}$. This parameter-free prediction emerges from modeling memory as associative search processes operating on random graphs defined by memory representation structure.

The theoretical foundation rests on two principles: memory items are represented by overlapping random sparse neuronal ensembles in dedicated memory networks, and recall follows associative chains where the next item recalled has the largest overlap with the current item. This associative retrieval mechanism provides mathematical validation for similarity-based recall processes, supporting computational approaches to memory assessment that rely on semantic similarity rather than exact matching.

2.2 | Note-Taking Techniques and Tools

Note-taking serves as an essential tool for facilitating memory retention and comprehension. Various techniques and tools have been developed to assist learners in organizing and processing information effectively.

2.2.1 | Traditional Note-Taking Techniques

Rashid and Rigas (2006) examine the comparative effectiveness of traditional note-taking methods, including the Cornell Method, the Outlining Method, and the Mapping Method. These methods provide learners with strategies for organizing content to improve comprehension and retention.

The Cornell note-taking method consists of three distinct sections: keywords area, notes area, and summary area, providing systematic organization without multiple recopying. Rashid and Rigas demonstrate its effectiveness across diverse subject domains through active interaction requirements that enable immediate review and improved comprehension via individual understanding during summarization processes. While alternative methods like outlining provide hierarchical organization, they require intensive concentration unsuitable for fast-paced lectures, and mapping methods demand active participation that may not be feasible in all educational contexts.

2.2.2 | Digital Note-Taking and E-Learning Tools

Rashid and Rigas (2006) further explore the role of technology in enhancing note-taking and memory retention within e-learning environments. The integration of tools such as Microsoft OneNote and Tablet PCs has provided learners with new ways to engage with content, enabling multimedia annotations and portable accessibility.

Rashid and Rigas identify seven critical parameters for effective e-learning environments: institutional support, course development, teaching integration, course structure, student support, faculty support, and evaluation frameworks. Digital note-taking technologies enable knowledge consolidation through multimedia integration where students combine textual notes with audio, visual, and interactive elements, supporting encoding phases of memory formation while providing searchable, editable, and shareable content.

The effectiveness depends on students' learning style classifications: divergers, assimilators, convergers, and accommodators, each requiring different technological approaches. The research demonstrates that effective digital tools enhance motivation while supporting academic achievement through appropriate technological integration that follows established learning models.

2.3 | Natural Language Processing for Semantic Similarity

Semantic similarity measurement represents a fundamental challenge in natural language processing, where the goal is to quantify the degree of meaning correspondence between textual units. This capability serves as the foundation for automated assessment systems that must evaluate student responses based on conceptual understanding rather than superficial text matching.

2.3.1 | Evolution of Semantic Similarity Approaches

Corley, Csomai, and Mihalcea (2005) establish the theoretical foundations for computational semantic similarity measurement, demonstrating that semantic relationships can be quantified through lexical analysis, semantic networks, and corpus-based statistical approaches. Their work identifies critical limitations in traditional approaches that rely primarily on lexical overlap, which fails to capture semantic relationships between synonymous expressions and conceptually equivalent phrasings.

The evolution from lexical to semantic approaches represents a fundamental shift in natural language understanding, where surface-level text matching gives way to meaning-based comparison. Corley et al. demonstrate that semantic similarity measurement must account for paraphrasing, conceptual equivalence, and contextual meaning variation to provide educationally meaningful assessment capabilities.

2.3.2 | Transformer Architecture and Contextualized Embeddings

Devlin et al. (2019) introduce BERT (Bidirectional Encoder Representations from Transformers), establishing the theoretical and practical foundations for contextualized language understanding. BERT's bidirectional training enables the model to consider both left and right context when generating word representations, addressing fundamental limitations of previous approaches that processed text unidirectionally.

The BERT architecture employs attention mechanisms that enable the model to selectively focus on relevant contextual information when generating representations. This attention-based approach proves particularly valuable for educational applications where word meaning depends critically on surrounding context. The model's ability to capture syntactic and semantic relationships through self-supervised learning eliminates the need for labeled similarity datasets while providing robust performance across diverse domains.

2.3.3 | Contrastive Learning for Sentence Embeddings

Gao, Yao, and Chen (2021) address critical limitations in applying pre-trained language models to sentence similarity tasks through Simple Contrastive learning of Sentence Embeddings (SimCSE). Their research reveals that standard BERT embeddings suffer from anisotropic distribution problems where sentence representations cluster in narrow regions of the vector space, artificially inflating similarity scores regardless of actual semantic relationships.

SimCSE solves this fundamental problem through contrastive learning that encourages semantically similar sentences to have similar embeddings while pushing dissimilar sentences apart. The key innovation involves using dropout as a data augmentation mechanism, where the same sentence processed with different dropout masks serves as positive pairs for contrastive training. This approach proves particularly valuable for educational applications where assessment systems must distinguish between genuine semantic similarity and superficial text overlap.

2.3.3.1 | Quizlet

Tran and Nguyen (2022) demonstrate that Quizlet integrates notes, spaced repetition, and active learning into an engaging platform, effectively improving vocabulary retention and long-term memory consolidation.

Tran and Nguyen's action research reveals that Quizlet application produces significant positive effects on students' vocabulary retention through systematic integration of short-term and long-term memory processes. The platform addresses critical gaps in traditional vocabulary instruction where students easily forget new words without extended practice, particularly in contexts where curriculum time constraints limit intensive vocabulary development.

The effectiveness of Quizlet stems from its implementation of word retention principles, defined as the ability to provide meaning of new words after extended time periods. The application facilitates knowledge consolidation by enabling students to engage with vocabulary both in classroom settings and during independent study, creating multiple exposure opportunities that strengthen memory encoding and retrieval pathways.

Quizlet's success demonstrates the importance of technology integration that follows established learning models. The platform's design aligns with the SAMR model (Substitution-Augmentation-Modification-Redefinition), enabling progressive technology integration that moves beyond simple substitution to meaningful educational transformation. Students report increased interest and engagement with vocabulary learning when using the application, indicating that effective digital tools can enhance motivation while supporting academic achievement.

2.3.4 | Multi-Dimensional Performance Assessment

Baars, Khare, and Ridderstap (2022) establish that effective mobile learning applications must integrate multiple assessment dimensions to provide comprehensive evaluation of student performance. Their exploration of students' use of mobile applications for self-regulated learning reveals that single-metric assessment fails to capture the complexity of learning processes, necessitating holistic approaches that consider various performance indicators.

2.4 | Progress Tracking and Gamification in Educational Technology

Progress tracking and gamification elements serve essential roles in maintaining student motivation and promoting consistent learning behaviors. These approaches leverage psychological principles of goal-setting, feedback, and achievement recognition to support sustained educational engagement.

2.4.1 | Gamification Theory and Educational Applications

Deterding et al. (2011) define gamification as "the use of game design elements in non-game contexts," distinguishing it from serious games and playful design through hierarchical typology ranging from interface patterns (badges, points) to deeper design principles. Effective gamification emphasizes structured, goal-driven experiences (gamefulness) rather than open-ended exploration (playfulness), requiring alignment with underlying user motivations to avoid overjustification effects where external rewards reduce intrinsic engagement, particularly critical in educational contexts where balanced extrinsic elements must maintain genuine learning satisfaction.

2.4.2 | Three-Star Rating Systems and Performance Feedback

Gaston and Cooper (2017) demonstrate three-star systems' effectiveness through Foldit protein folding game analysis, revealing that players exposed to three-star ratings completed levels with fewer moves and increased deliberation, categorizing these as "Rewards of Glory" that provide psychological satisfaction without affecting gameplay mechanics. The research shows three-star systems encourage mastery-oriented behavior through level replay (approximately twice as often as control groups) and appeal primarily to Achievers, though they don't significantly impact long-term retention, supporting implementation of structured failure states that encourage learning from mistakes.

2.4.3 | Mobile Learning Applications and Habit Formation

Stojanovic, Grund, and Fries (2020) demonstrate that app-based habit building systems effectively reduce motivational impairments during studying through structured progress feedback and consistency tracking, identifying key factors: specific goal setting, immediate feedback, progress visualization, and consistency rewards that work synergistically to support long-term engagement beyond initial motivation phases.

Their research establishes that mobile learning applications achieve maximum effectiveness when integrated seamlessly into daily routines while providing meaningful learning experiences, enabling frequent, brief sessions that accumulate into substantial educational progress while supporting spaced learning principles that optimize memory retention.

2.4.4 | Motivational Design in Educational Technology

Vișcu (2024) provides comprehensive analysis of optimizing educational environments for active learning methodologies, establishing that effective educational technology must balance challenge and support while addressing three key motivational factors: autonomy (student control), competence (achievable yet challenging goals), and relatedness (meaningful outcomes). The research emphasizes that immediate feedback systems prove most effective when providing actionable information for performance improvement rather than merely documenting achievement, distinguishing formative from summative feedback approaches in educational technology design.

Vișcu's analysis reveals that immediate feedback systems prove most effective when they provide actionable information that students can use to improve their performance rather than merely documenting their current achievement level. This distinction emphasizes the importance of formative rather than summative feedback approaches in educational technology design.

2.4.4.1 | ChatGPT and the Generation of Educational Content

Mishra and Anthony (2023) highlight both the benefits and drawbacks of AI tools like ChatGPT in education, emphasizing the need for responsible integration to enhance critical thinking and active learning processes.

Mishra and Anthony's comprehensive analysis reveals that ChatGPT usage among college students is driven by performance anticipation, expected effort reduction, and perceived educational value, with college students representing approximately 33.1

The study demonstrates that ChatGPT adoption follows established technology acceptance models, where perceived usefulness and ease of use predict likelihood of continued usage. However, the research also reveals concerning dependency patterns where students may rely excessively on AI-generated content rather than developing independent critical thinking and note-taking skills. This dependency can undermine the knowledge consolidation process by reducing active engagement with learning materials.

For educational applications like note-taking and recall systems, understanding behavioral intention toward AI tools becomes critical. Students' motivation to use AI assistance must be balanced with maintaining personal agency in learning processes. Effective integration requires designing systems that leverage AI capabilities while preserving the cognitive benefits of active note-taking, personal reflection, and knowledge consolidation that traditional learning processes provide.

2.4.4.2 | Self-Regulated Learning by Using Study Applications

Baars, Khare, and Ridderstap (2022) demonstrate that study applications with well-designed user interfaces support self-regulated learning by enabling goal-setting, time management, and feedback during study sessions.

2.5 | NLP and Semantic Similarity

The development of NLP techniques has significantly advanced models that analyze and quantify semantic similarity.

2.5.1 | Word Embedding Foundations

The evolution of neural word embeddings represents a foundational shift in computational linguistics, establishing the vector space principles that underpin contemporary sentence embedding methodologies. Church (2017) demonstrates that Word2Vec's widespread adoption stems not from theoretical complexity, but from its practical accessibility and the availability of downloadable implementations that enable immediate application. The method's core innovation lies in representing words as dense vectors in continuous space, where semantic relationships are preserved through geometric relationships rather than symbolic representations.

Mikolov et al. (2013) introduced the skip-gram architecture that learns word representations by predicting contextual words, establishing contrastive learning principles that would later be adapted for sentence-level embeddings. The famous analogy $man : woman :: king : queen$ illustrates how vector arithmetic captures semantic relationships, with the mathematical operation $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$ demonstrating the foundation of semantic similarity computation through vector mathematics.

Jatnika, Bijaksana, and Suryani (2019) provide empirical validation of Word2Vec's semantic similarity capabilities through extensive evaluation on WordSim-353 and SimLex-999 datasets. Their analysis reveals that cosine similarity, computed as the angular distance between word vectors, effectively quantifies semantic relationships with correlation scores reaching 0.665 on WordSim-353. This establishes the mathematical foundation for similarity measurement that extends to sentence-level embeddings, where the principle that semantically similar entities should be positioned closer in vector space forms the theoretical basis for contrastive learning objectives used in SimCSE and similar frameworks.

2.5.2 | Text Semantic Similarity

Corley et al. (2005) provided foundational insights into computational methods for evaluating semantic similarity, focusing on techniques that measure the closeness of meaning between texts.

2.5.3 | The BERT Framework

Devlin et al. (2019) emphasize that BERT leverages bidirectional representations of language to capture semantic relationships through its transformer-based architecture.

2.5.3.1 | SimCSE

Gao, Yao, and Chen (2021) developed SimCSE, a fine-tuning approach that enhances the quality of sentence embeddings, making them suitable for tasks like semantic similarity evaluation and textual entailment.

SimCSE addresses a fundamental challenge in sentence embeddings known as representation collapse, where different sentences produce nearly identical vector representations, rendering similarity measurements meaningless. Traditional BERT embeddings suffer from this phenomenon due to anisotropic distribution of representations in vector space, where embeddings cluster in narrow regions rather than utilizing the full dimensional space effectively.

The theoretical contribution of SimCSE lies in transforming anisotropic embedding distributions into more isotropic ones. Anisotropic embeddings concentrate in narrow cones within the vector space, limiting their ability to capture fine-grained semantic distinctions necessary for educational assessment. The contrastive learning framework regularizes the embedding space to achieve uniform distribution, where sentences with different meanings occupy distinct regions while semantically similar sentences remain proximate.

Unlike previous approaches requiring complex data augmentation techniques such as back-translation, paraphrasing, or syntactic transformations, SimCSE achieves superior performance using only dropout noise as minimal data augmentation. This approach preserves semantic content while introducing controlled perturbations necessary for contrastive learning. When the same sentence passes through the encoder twice with different dropout masks, the resulting embeddings form natural positive pairs without external modification.

The mathematical foundation of SimCSE rests on achieving uniform distribution property in the embedding space. This property ensures that the similarity function can effectively discriminate between sentences with varying degrees of semantic overlap, eliminating the bias present in anisotropic embeddings where high similarity scores may result from distributional artifacts rather than genuine semantic relationships.

2.5.3.2 | Contrastive Learning

SimCSE's unsupervised framework generates "positive pairs" by applying independent dropout masks to the same input sentence. Specifically, when a sentence is passed through the BERT encoder twice, the randomness introduced by dropout produces two slightly different embeddings.

SimCSE's contrastive learning framework builds upon foundational principles established in Word2Vec's skip-gram model while extending them to sentence-level representations. The approach creates positive pairs through minimal data augmentation and efficiently generates negative samples through in-batch negatives, eliminating the computational overhead of explicitly creating negative pairs while ensuring diverse negative samples.

For a given sentence s_i, all other sentences s_j where $j \neq i$ in the batch serve as negatives, creating rich contrastive signals without additional data preprocessing. This in-batch negative sampling strategy demonstrates continuity with Word2Vec's contrastive learning principles, where the fundamental approach of learning through contrast remains consistent across word-level and sentence-level embeddings.

$$\ell_i = -\log \frac{\exp(\mathrm{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(h_i, h_j)/\tau)}$$

Figure 2.1: Formula for Contrastive Learning

2.5.3.3 | Cosine Similarity

Cosine similarity quantifies the angular distance between embeddings, ensuring that semantically similar sentences are pulled closer together, while dissimilar ones are pushed apart. The temperature parameter τ sharpens the similarity distribution, controlling the influence of positive and negative pairs during training.

The temperature parameter τ in the contrastive learning formula plays a critical role in controlling the sharpness of the similarity distribution. Lower temperature values create more peaked distributions, emphasizing high-confidence similarities, while higher values produce smoother distributions. The optimal temperature balances discriminative power with stability, preventing over-confident predictions that could lead to poor generalization in educational assessment contexts.

$$\mathrm{sim}(h_1, h_2) = \frac{h_1^\top h_2}{\|h_1\| \|h_2\|}$$

Figure 2.2: Formula for Cosine Similarity

2.6 | Progress Tracking

Progress tracking refers to systems that enable users to monitor and evaluate their advancement toward specific goals. This section details the existing research on digital methods of tracking user progress, and the idea of game design elements (that resemble progress tracking) being used outside its usual context as a self-driving motivational tool..

2.6.1 | Progress Tracking as Part of Gameful Design

Deterding et al. (2011) emphasize the importance of structured systems that guide users through progressive goals and stages of development, helping maintain motivation and engagement.

Deterding et al. establish gamification as "the use of game design elements in non-game contexts," providing a precise conceptual framework that distinguishes gamification from serious games, playful design, and other related concepts. This definition enables systematic analysis of progress tracking systems through their component game design elements rather than relying on vague notions of "making things fun" or general engagement enhancement.

The hierarchical typology of game design elements proposed by Deterding et al. ranges from interface design patterns (badges, points, leaderboards) to deeper game design principles and methodologies. This classification system supports the distinction between shallow gamification, which merely adds superficial UI elements, and deeper gamification that incorporates structural design principles for meaningful behavioral change.

Gamefulness, as distinct from playfulness, emphasizes structured, goal-driven experiences that mirror the emotional and experiential qualities of games rather than open-ended exploration. This distinction is crucial for educational progress tracking systems, where structured advancement through defined objectives supports learning outcomes more effectively than unstructured play activities.

The conceptual framework demonstrates that effective gamification systems must import formal qualities of game design; including clear goals, rules, and feedback mechanisms; into educational contexts to encourage sustained engagement while maintaining focus on learning objectives rather than entertainment value.

2.6.1.1 | The Three-Star Scoring System

Gastón and Cooper (2017) explore the efficacy of the three-star scoring system, highlighting its ability to improve user engagement and performance in educational and gamified applications.

Gaston and Cooper's controlled experiment with the protein folding game Foldit reveals that three-star systems effectively encourage more thoughtful play during onboarding phases. Their research demonstrates that players exposed to three-star rating systems completed levels with fewer extra moves and took more time per move, indicating increased deliberation and strategic thinking during task completion.

The study identifies three-star systems as "Rewards of Glory"; purely cosmetic rewards that do not affect gameplay mechanics but provide psychological satisfaction through achievement recognition. This categorization is particularly relevant for educational applications where intrinsic motivation and mastery recognition matter more than extrinsic gameplay advantages.

Critically, while three-star systems improved behavioral engagement metrics; specifically encouraging desired behaviors like careful execution and level replay; they did not significantly impact long-term retention rates. Players in three-star conditions replayed completed levels approximately twice as often as control groups, suggesting that star-based feedback systems promote mastery-oriented behavior and knowledge consolidation through repetitive practice.

The research reveals important player type considerations, with three-star systems primarily appealing to Achievers (as categorized by Bartle's player taxonomy) who are motivated by completing tasks and overcoming challenges. This finding suggests that effective progress tracking systems must accommodate diverse motivational profiles while providing meaningful feedback for performance improvement.

The forced reset variant of the three-star system, where players were required to restart levels after exceeding move thresholds, further amplified careful play behaviors. This finding supports the implementation of structured failure states that encourage learning from mistakes rather than allowing progression through suboptimal performance.

2.7 | Summary

The studies reviewed in this chapter highlight key advancements in note-taking strategies, memory processes, NLP, and progress tracking. These insights establish a foundation for the proposed system, integrating effective learning strategies and semantic similarity evaluation to optimize recall and knowledge retention.

The table compares various studies based on their focus on specific aspects or features that are relevant to the proposed approach. It highlights which aspects are addressed and identifies gaps that the proposed approach aims to fill by integrating these features into a unified framework.

Table 2.1: Comparison Table of Related Literature

Aspects/ Features:	**Integration of Semantic Similarity Models**	**Progress Tracking**	**Focus on Knowledge Consolidation, Memory, and Recall**	**Flexibility for Varied Note Structures (Formatting)**	**Feedback and Performance Metrics**	**App with Interactive UI for Enhanced Usability**	**Use of NLP for Evaluating Semantic Understanding**
Studies:							
Stojanovic et al., 2020	×	×	×	×	×	✓	×
Tran et al., 2020	×	✓	✓	×	✓	✓	×
Baars et al., 2003	×	×	✓	×	×	✓	×
Proposed Approach	✓	✓	✓	✓	✓	✓	✓

3

Theoretical Framework

The theoretical framework for the Consol system draws from multiple interconnected domains of research that provide the scientific foundation for semantic similarity-based learning assessment. This framework integrates memory and cognitive science research with natural language processing theory and progress tracking methodologies to establish a cohesive foundation for the Consol platform.

The framework encompasses four primary theoretical domains: cognitive memory processes that explain how students naturally encode and retrieve information (Sachs, 1967; Karpicke & Roediger, 2008; Naim et al., 2020), semantic similarity measurement principles that enable computational assessment of meaning relationships (Gao et al., 2021; Devlin et al., 2019), progress tracking and motivation theories that support sustained learning engagement (Deterding et al., 2011; Gaston & Cooper, 2017), and learning analytics frameworks that provide educational insight through data visualization (Baars et al., 2022; Stojanovic et al., 2020).

These theoretical foundations are laid out in a matrix in Figure **??** which demonstrates how each research domain supports specific aspects of the Consol platform, along with its variables and methods used, and again depicted as a diagram in Figure 3.1, ensuring that system design decisions align with established educational and computational theory. The theoretical architecture diagram (Figure 3.1) provides a visualization of how these theories integrate to support Consol's functionality by also tying each researched aspect into a tangible feature that ends up in Consol's web-based platform.

3.1 | Theoretical Research Matrix

The theoretical foundation of Consol rests upon systematic integration of research from multiple domains. Table 3.1 presents the comprehensive mapping of theoretical foundations to specific platform capabilities, demonstrating how each research study contributes to the system's educational effectiveness.

Table 3.1: Theoretical Research Matrix showing mapping of literature to Consol platform capabilities

Authors	Focus Variables	Independent Variables	Other Variables	Theoretical Framework	Method Used
Gao et al. (2021)	Sentence embeddings, Semantic similarity	Contrastive learning, Dropout masks, BERT encoder	Training data, Model parameters	Contrastive Learning Theory	Experimental validation
Sachs (1967)	Memory recognition, Semantic vs syntactic	Time intervals, Material interpolation	Sentence complexity, Participant variability	Memory Consolidation Theory	Experimental psychology
Karpicke & Roediger (2008)	Retrieval practice, Long-term retention	Testing frequency, Study repetition	Material difficulty, Participant motivation	Retrieval Practice Theory	Experimental design
Deterding et al. (2011)	Game design elements, Educational motivation	Gamification mechanics, Feedback systems	User engagement, Task complexity	Gamification Theory	Theoretical analysis
Gaston & Cooper (2017)	Three-star rating, Performance feedback	Rating thresholds, Achievement levels	User behavior, Task completion	Performance Feedback Theory	Game analytics
Naim et al. (2020)	Memory recall, Mathematical models	List length, Presentation rate	Cognitive capacity, Individual differences	Mathematical Memory Theory	Statistical modeling
Baars et al. (2022)	Mobile learning, Self-regulation	Application features, Usage patterns	Learning outcomes, Technology acceptance	Mobile Learning Theory	Mixed methods
Klemm (2007)	Working memory, Learning efficiency	Memory capacity, Cognitive load	Academic performance, Individual ability	Memory-Learning Integration	Theoretical review

3.2 | Cognitive Science and Memory Theory Foundations

The cognitive foundations of the Consol platform rest on established memory research that explains how students naturally encode, consolidate, and retrieve information. These theoretical principles provide the scientific basis for understanding why semantic similarity-based assessment aligns with natural cognitive processes.

3.2.1 | Working Memory and Learning Integration Theory

Klemm (2007) establishes working memory as the operational foundation for intellectual performance, functioning as a cognitive workspace where information chunks undergo processing during thinking operations. The research demonstrates tight correlations between working memory capacity, intellectual ability, and problem-solving performance, revealing that students' learning effectiveness depends fundamentally on their ability to maintain and manipulate information during cognitive tasks.

This theoretical foundation supports Consol's design philosophy that accommodates natural memory processes rather than imposing artificial constraints. The platform provides an environment where students can observe their own memory performance patterns through similarity tracking and progress visualization, enabling them to identify whether their recall demonstrates improvement, stability, or decline over time without external judgment of their cognitive capabilities.

3.2.2 | Retrieval Practice Theory

Karpicke and Roediger (2008) demonstrate that retrieval practice constitutes a fundamental mechanism for long-term memory consolidation, with testing effects producing superior retention compared to repeated study. Their systematic investigation reveals that once students initially learn material, continued studying provides minimal retention benefits, whereas continued testing generates substantial positive effects on memory consolidation.

The research establishes that students exhibit poor metacognitive awareness of effective learning strategies, with performance predictions showing no correlation with actual retention outcomes. This metacognitive blindness suggests that students may persist with ineffective study methods while avoiding retrieval practice that would substantially improve learning outcomes. Consol accommodates this research by providing a platform where students can engage in systematic retrieval practice while observing their own performance patterns, enabling them to discover the effectiveness of active recall through direct experience rather than external instruction.

3.2.3 | Semantic Memory Prioritization Theory

Sachs (1967) provides foundational evidence that memory systems exhibit temporal dynamics where semantic information persists while syntactic details rapidly decay. The research demonstrates that after minimal interpolated material (80-160 syllables), recognition for syntactic changes drops to chance levels while semantic recognition remains robust, revealing that individuals naturally retain meaning rather than exact form.

This temporal separation of semantic and syntactic memory provides theoretical justification for similarity-based evaluation systems that align with natural memory processes. Consol's semantic similarity assessment accommodates the natural tendency toward meaning preservation while maintaining academic rigor, enabling students to observe whether their recall maintains semantic fidelity regardless of syntactic variation.

3.2.4 | Mathematical Memory Laws Theory

Naim et al. (2020) establish that human memory recall follows universal mathematical laws derived from first principles, with memory operating as deterministic processes rather than random phenomena. Their research on free recall paradigms reveals that recall performance follows predictable mathematical relationships based on associative network structures where memory items exist as overlapping neuronal ensembles.

The mathematical foundation demonstrates that retrieval occurs through associative chains where subsequent items are accessed based on similarity to currently active items. This theoretical framework supports Consol's implementation of semantic similarity measurement as an assessment mechanism, providing students with quantitative feedback that reflects the mathematical principles underlying their own memory retrieval processes.

3.3 | Semantic Similarity and Natural Language Processing Theory

The computational assessment of semantic similarity forms the technical core of the Consol platform, requiring sophisticated natural language processing approaches that can distinguish between genuine semantic relationships and superficial textual overlap. The theoretical foundation draws from advances in transformer architectures, contrastive learning principles, and vector space semantics.

3.3.1 | Transformer Architecture and BERT Foundations

Devlin et al. (2019) establish the transformer architecture as a breakthrough in natural language understanding through bidirectional encoder representations. BERT's key innovation lies in training on masked language modeling and next sentence prediction tasks that force the model to develop comprehensive understanding of linguistic relationships rather than relying on surface-level patterns.

The theoretical significance extends beyond technical improvement to fundamental changes in how machines process language. BERT's bidirectional attention mechanism enables each token to attend to all other tokens in the sequence, creating rich contextual representations where word meanings depend on complete sentence context. This architectural foundation provides the embedding space within which semantic similarity measurements operate, establishing vector representations that capture nuanced meaning relationships essential for educational assessment.

3.3.2 | SimCSE Contrastive Learning Theory and Technical Implementation

Gao, Yao, and Chen (2021) address the fundamental challenge that while BERT produces excellent word-level representations, direct averaging or pooling of token embeddings yields poor sentence-level similarity measurements. The anisotropic distribution problem occurs where sentence vectors cluster in narrow regions of the vector space, with high cosine similarity scores between semantically unrelated sentences reaching 0.6-0.8 regardless of actual meaning relationships.

SimCSE introduces contrastive learning specifically designed for sentence embedding optimization through unsupervised learning approaches. The unsupervised method leverages dropout as implicit data augmentation, treating the same input sentence with different dropout masks as positive pairs, enabling effective learning without requiring labeled training data.

The core innovation lies in the training objective that maximizes agreement between augmented views of identical sentences while minimizing agreement between different sentences. For unsupervised SimCSE, given a sentence x_i and its corresponding embedding $h_i = f_\theta(x_i)$ where f_θ represents a pre-trained language model, the training loss function is:

$$\ell_i = -\log \frac{e^{\mathrm{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(h_i, h_j^+)/\tau}} \tag{3.1}$$

where h_i^+ represents the same sentence with different dropout applied, $\mathrm{sim}(\cdot, \cdot)$ computes cosine similarity, τ is the temperature parameter controlling distribution sharpness, and N represents batch size.

The unsupervised approach demonstrates effectiveness through dropout-based augmentation that creates meaningful positive pairs from identical sentences. This method enables robust semantic representation learning without requiring external labeled datasets, making it particularly suitable for educational applications where domain-specific training data may be limited.

3.3.3 | Dropout-Based Positive Sampling Theory

The theoretical breakthrough of SimCSE lies in recognizing dropout as an effective positive sampling strategy that eliminates the need for manual data augmentation while maintaining semantic consistency. For each input sentence x_i, the framework generates positive pairs through different dropout masks applied to the same encoder. The same sentence processed with different dropout patterns creates subtle variations in hidden states while preserving semantic content.

This approach forces the model to learn robust representations that capture meaning rather than relying on surface-level patterns that might be randomly masked. The different dropout masks create variations in attention weights and hidden representations, teaching the model to focus on meaning-preserving features that remain consistent across different dropout realizations.

3.3.4 | Uniform Distribution Property and Anisotropy Solutions

SimCSE addresses the anisotropy problem by encouraging uniform distribution of sentence embeddings in vector space through contrastive learning. The training objective pushes semantically different sentences away from each other while pulling similar sentences together, creating a more uniform distribution that utilizes the full dimensional capacity of the embedding space.

The uniform distribution property ensures that high similarity scores indicate genuine semantic relationships rather than distributional artifacts. This addresses the critical limitation of standard BERT sentence embeddings where random sentences often achieve similarity scores above 0.5 despite having no meaningful semantic relationship.

3.3.5 | Consol's Implementation of SimCSE Assessment

Consol implements unsupervised SimCSE models based on BERT pre-training to ensure robust semantic similarity measurement for educational assessment. The platform utilizes contrastive learning principles that align with natural memory processes, providing assessment that reflects genuine semantic understanding rather than superficial textual matching.

The three-tier threshold system directly applies SimCSE similarity scores: scores above 0.8 indicate comprehensive semantic alignment where student responses demonstrate mastery-level understanding equivalent to paraphrasing or detailed explanation, scores between 0.6-0.8 suggest adequate comprehension with minor conceptual gaps, and scores between 0.3-0.6 indicate partial understanding requiring additional study. This threshold calibration reflects SimCSE's demonstrated correlation with human semantic similarity judgments across educational contexts.

The immediate feedback mechanism provides real-time similarity scores that enable students to observe their recall precision while the memory retrieval process remains active. This design accommodates Karpicke and Roediger's (2008) findings about retrieval practice effectiveness by allowing students to immediately assess whether their recall maintains semantic fidelity to the original material.

3.3.6 | Semantic Assessment Integration with Consol Note Management

Consol's note management system leverages SimCSE semantic similarity to support the natural progression from initial note creation through recall assessment. Students create notes using either manual input through the note editor interface or file upload functionality, establishing a knowledge base that serves as the semantic reference for subsequent recall sessions.

The note creation interface accommodates diverse content types while maintaining focus on meaningful information capture rather than formatting constraints. This approach aligns with Sachs' (1967) findings about memory's preference for semantic content over syntactic structure, enabling students to focus on conceptual understanding rather than exact reproduction requirements.

During recall sessions, the platform compares student-generated responses against their original notes using SimCSE similarity calculation. The immediate similarity feedback enables students to assess whether their recall maintains semantic fidelity to their original understanding, supporting the retrieval practice principles established by Karpicke and Roediger (2008) through objective performance measurement.

3.3.7 | Three-Star Rating System Implementation

Consol implements the three-star rating system directly based on Gaston and Cooper's (2017) research demonstrating that three-tier performance indicators optimize motivation while maintaining meaningful differentiation. The star allocation follows graduated similarity thresholds: three stars for scores above 0.8 (comprehensive semantic alignment), two stars for scores between 0.6-0.8 (adequate comprehension), one star for scores between 0.3-0.6 (partial understanding), and zero stars for scores below 0.3 (insufficient recall).

These thresholds reflect SimCSE's calibrated performance on semantic similarity benchmarks, where scores above 0.8 correspond to human judgments of semantic equivalence, and scores below 0.3 indicate semantic independence. The three-star system provides immediate performance feedback that students can interpret without requiring technical understanding of similarity measurement algorithms.

The daily streak calendar visualizes consistency performance by tracking consecutive days with successful note recall sessions, implementing the habit formation principles established by Stojanovic et al. (2020). The calendar interface displays the highest star rating achieved each day, enabling students to observe their learning consistency patterns while providing motivation for sustained engagement through visible progress accumulation.

3.4 | Progress Tracking and Motivation Theory

The sustained engagement required for effective learning necessitates systematic progress tracking and motivational design that supports long-term educational commitment. The theoretical foundation draws from gamification research, habit formation studies, and motivational psychology to create environments that encourage consistent learning behaviors.

3.4.1 | Mobile Learning and Self-Regulation Theory

Baars et al. (2022) establish that effective mobile learning environments must integrate multiple assessment dimensions to provide comprehensive evaluation of student performance. Their exploration of students' use of mobile technology for self-regulated learning reveals that single-metric assessment fails to capture the complexity of learning processes, necessitating holistic approaches that consider various performance indicators including engagement patterns, consistency measures, and improvement trends.

The theoretical framework emphasizes that learning platforms (not necessarily just mobile) achieve maximum effectiveness when integrated seamlessly into daily routines while providing meaningful learning experiences. This integration enables frequent, brief sessions that accumulate into substantial educational progress while supporting spaced learning principles that optimize memory retention through distributed practice. This materializes as Consol's daily streak feature, where consistency is visually tracked to encourage habitual engagement without imposing rigid usage requirements that could undermine intrinsic motivation. The minimum requirement of one session per day is enough stimulate the learning process, and the self-regulated goal to improve one's own performance is what drives continued usage, and therefore steady progress in memory consolidation.

3.4.2 | Gamification Theory in Educational Contexts

Deterding et al. (2011) establish gamification as "the use of game design elements in non-game contexts," providing theoretical distinction from serious games and playful design through hierarchical typology ranging from interface patterns to deeper design principles. Effective gamification emphasizes structured, goal-driven experiences (gamefulness) rather than open-ended exploration (playfulness), requiring careful alignment with underlying user motivations to avoid overjustification effects where external rewards reduce intrinsic engagement.

The theoretical framework recognizes that educational gamification must maintain focus on learning objectives while incorporating motivational elements that support rather than supplant genuine educational satisfaction. Consol accommodates this balance by providing progress visualization and achievement recognition that enables students to observe their own learning patterns without creating dependency on external validation for continued engagement.

3.4.3 | App-Based Habit Formation Theory

Stojanovic, Grund, and Fries (2020) establish that app-based habit building systems effectively reduce motivational impairments during studying through structured progress feedback and consistency tracking. Their research identifies key theoretical factors: specific goal setting that provides clear learning targets, immediate feedback that confirms progress toward objectives, progress visualization that makes improvement visible over time, and consistency rewards that recognize sustained engagement patterns.

These factors work synergistically to support long-term engagement beyond initial motivation phases, enabling learners to maintain educational practices even when immediate enthusiasm wanes. Consol accommodates these principles by providing an environment where students can observe their own study consistency patterns and improvement trends without external pressure to maintain artificial performance standards.

3.4.4 | Immediate Feedback Implementation in Consol

Karpicke and Roediger (2008) establish that feedback timing significantly impacts learning effectiveness, with immediate feedback promoting better knowledge consolidation than delayed feedback. The Consol system provides immediate similarity feedback that enables students to assess their recall accuracy in real-time during the note recollection process.

The immediate feedback mechanism displays similarity scores as students complete their recall attempts, enabling them to understand the quality of their memory retrieval while the cognitive processes remain active. This real-time assessment supports metacognitive awareness by helping students identify conceptual gaps and adjust their understanding during the learning session rather than waiting for delayed evaluation.

3.5 | Automated Assessment Validity Theory

A valid automated assessment requires consistency in measurement across equivalent inputs, sensitivity to meaningful differences in understanding quality, objectivity through elimination of human scorer bias, and educational relevance through alignment with learning objectives. SimCSE-based assessment satisfies these criteria by providing deterministic similarity scores that reflect semantic relationships while maintaining objectivity across diverse response styles.

In this study, this implies that Consol having a consistent and deterministic scoring agent prevents situations such as arbitrary scoring, which is undeniably prevalent in human interventions. It is unrealistic for human-like logic to constantly affirm similarity by arriving at a certain number without injecting a degree of subjectivity or personal bias. Automated assessment, especially from the likes of SimCSE, would not be susceptible to fluctuations in scoring. In return, this would provide students with a more reliable and consistent feedback mechanism that they can trust to accurately reflect their learning progress.

Confidence in the assessment is foundational for the rest of the progress tracking and motivation theories, as well as its implementations.

3.5.1 | Data Visualization and Learning Analytics

Building upon the foundation of reliable automated assessment, effective educational data visualization requires systematic collection and analysis of learning data to provide enhanced educational understanding through quantitative insights into learning processes. Baars et al. (2022) establish that effective learning analytics require multi-dimensional assessment approaches that capture learning complexity beyond single metrics, emphasizing the importance of comprehensive performance indicators including engagement patterns, consistency measures, and improvement trends.

Vișcu (2024) provides comprehensive analysis of optimizing educational environments for active learning methodologies, establishing that effective educational technology must balance challenge and support while addressing three key motivational factors: autonomy (student control over learning processes), competence (achievable yet challenging goals), and relatedness (meaningful outcomes connected to personal objectives). The research emphasizes that immediate feedback systems prove most effective when providing actionable information that students can use to improve their performance rather than merely documenting their current achievement level.

This integrated foundation encompasses data visualization principles supported by educational technology research (Vișcu, 2024), self-regulated learning theory that emphasizes student autonomy in monitoring their own progress (Baars et al., 2022), and educational measurement approaches that support rather than replace human judgment in learning assessment. Stojanovic et al. (2020) demonstrate that systematic progress tracking through app-based systems effectively reduces motivational impairments during studying by providing structured feedback and consistency visualization that enables learners to maintain educational practices beyond initial motivation phases.

The distinction between formative and summative feedback approaches proves critical in educational technology design, enabling students to make informed decisions about their learning strategies based on objective performance data derived from the reliable assessment foundation established through automated scoring consistency.

3.5.2 | Multi-Dimensional Performance Assessment

Effective educational assessment requires multiple performance indicators that capture different aspects of learning achievement. Consol implements multi-dimensional measurement through the radar chart visualization that displays three primary metrics: comprehension (average semantic similarity across sessions), speed (words per minute during recall), and mastery (percentage of sessions achieving three-star performance).

This three-dimensional assessment recognizes that learning involves multiple cognitive processes that cannot be captured through single metrics. Students achieving high comprehension with moderate speed demonstrate thorough understanding that prioritizes accuracy over efficiency, while balanced performance across all three dimensions indicates comprehensive learning development. The session metadata provides additional granular analysis including word count tracking and attempt frequency measurement.

3.5.3 | Analytics Dashboard and Three-Dimensional Performance Measurement

Consol's analytics dashboard implements multi-dimensional performance assessment through radar chart visualization that displays three primary learning metrics: Comprehension (semantic similarity performance), Speed (session completion efficiency), and Mastery (consistency of high-performance recall). This three-axis system directly applies Baars et al.'s (2022) findings about mobile learning requiring holistic assessment approaches that capture learning complexity beyond single metrics.

The radar chart visualization provides immediate insight into student learning patterns, enabling identification of balanced development versus specialized strengths. Students achieving high comprehension with moderate speed demonstrate thorough understanding that prioritizes accuracy over efficiency, while high speed with adequate comprehension suggests developing fluency that benefits from continued practice. The mastery dimension reflects consistency across multiple sessions, indicating knowledge consolidation over time.

Session metadata panels provide detailed performance breakdowns including word count analysis, attempt tracking, and temporal performance measurement. This granular data supports the mathematical memory principles established by Naim et al. (2020), enabling students to observe how their recall performance follows predictable patterns based on material complexity and practice frequency.

3.6 | Integrative Framework

The integration of cognitive science, computational linguistics, gamification, and learning analytics theories creates a comprehensive framework that supports specific Consol platform features through direct theoretical application rather than general design principles.

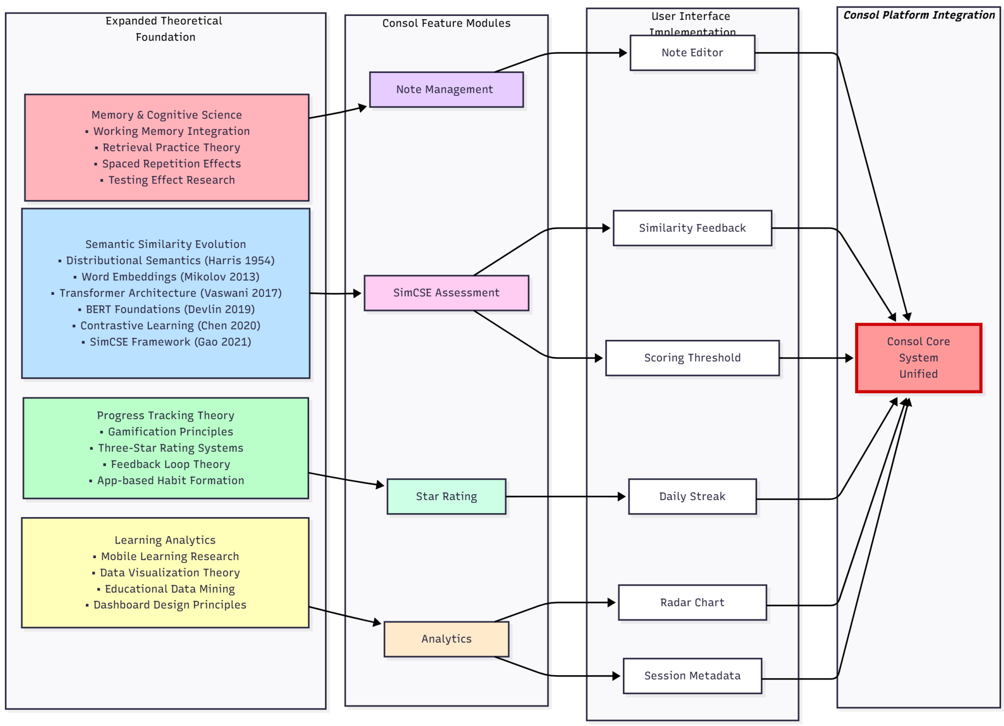

Figure 3.1: Comprehensive integrative framework showing the evolution from foundational research domains through Consol feature implementation to final platform integration. The diagram illustrates how theories from memory science, semantic similarity research, progress tracking, and learning analytics converge into Consol's unified system.

3.6.1 | Direct Theory-to-Feature Implementation

Memory consolidation theory (Karpicke & Roediger, 2008) directly supports Consol's note recall session structure where students recreate their notes from memory, engaging in retrieval practice that strengthens long-term retention. The platform's immediate similarity feedback provides students with objective assessment of their recall accuracy, supporting the metacognitive awareness that Karpicke and Roediger identify as crucial for effective learning.

SimCSE semantic similarity theory (Gao et al., 2021) enables the computational assessment engine that evaluates student recall against original notes. The arbitrarily-adjusted three-tier threshold system translates SimCSE similarity scores into meaningful educational feedback through the star rating display, providing students with immediate understanding of their performance level.

Progress tracking theory (Deterding et al., 2011) supports the daily streak calendar that tracks consecutive study days, encouraging habit formation without creating dependency on external rewards. The calendar visualization enables students to observe their consistency patterns while maintaining focus on learning rather than competition.

Learning analytics theory (Baars et al., 2022) supports the multi-dimensional radar chart that displays comprehension, speed, and mastery metrics. This visualization provides students with comprehensive performance insight that captures learning complexity beyond single similarity scores, enabling identification of balanced development versus specialized strengths.

3.7 | Summary

The theoretical framework establishes a comprehensive foundation for the Consol system by integrating cognitive science, natural language processing, and educational technology principles. The convergence of these theoretical domains creates a robust justification for semantic similarity-based educational assessment that aligns with natural memory processes while providing objective, actionable feedback.

Memory consolidation theory provides the conceptual foundation, establishing that learning involves progressive abstraction where students naturally compress and restructure information while preserving essential meaning. This principle directly supports assessment approaches that recognize semantic equivalence rather than demanding exact reproduction, positioning the Consol system as educationally sound rather than technologically convenient.

SimCSE contrastive learning theory offers the computational foundation, providing mathematically rigorous methods for measuring semantic similarity that aligns just enough with human cognitive processes, but not entirely that it becomes subjective and flimsy in providing similarity scores. The framework's ability to capture semantic relationships through vector space geometry enables objective assessment that reflects genuine understanding while maintaining consistency across diverse response styles.

Progress tracking and motivation theory, accompanied by the learning analytics theory provide the developmental principles for user interface design and feedback mechanisms that support learning effectiveness. The integration of progress tracking with immediate feedback creates an environment that promotes consistent practice while providing meaningful performance measurement.

Together, these theoretical foundations justify the Consol system as a principled approach to self-learning technology that bridges cognitive science insights with practical assessment needs. The system represents not merely a technical implementation as a proof-of-concept, but a theoretically grounded intervention that addresses fundamental challenges in learning assessment based on key pillars of memory research, and educational psychology.

4

Methodology

4.1 | System Architecture Design

The Consol system architecture integrates semantic similarity evaluation with user-centered interface design to create an effective recall assessment platform. Following established patterns in educational technology development, the architecture emphasizes modularity, scalability, and pedagogical effectiveness while ensuring seamless integration between natural language processing capabilities and user interaction requirements.

4.1.1 | Basic System Architecture

The fundamental architecture of the Consol system follows a layered approach that facilitates clear separation of concerns while maintaining efficient data flow throughout the educational assessment process. Figure 4.1 presents the high-level system architecture, illustrating the complete data flow cycle from user input through semantic analysis to performance feedback.

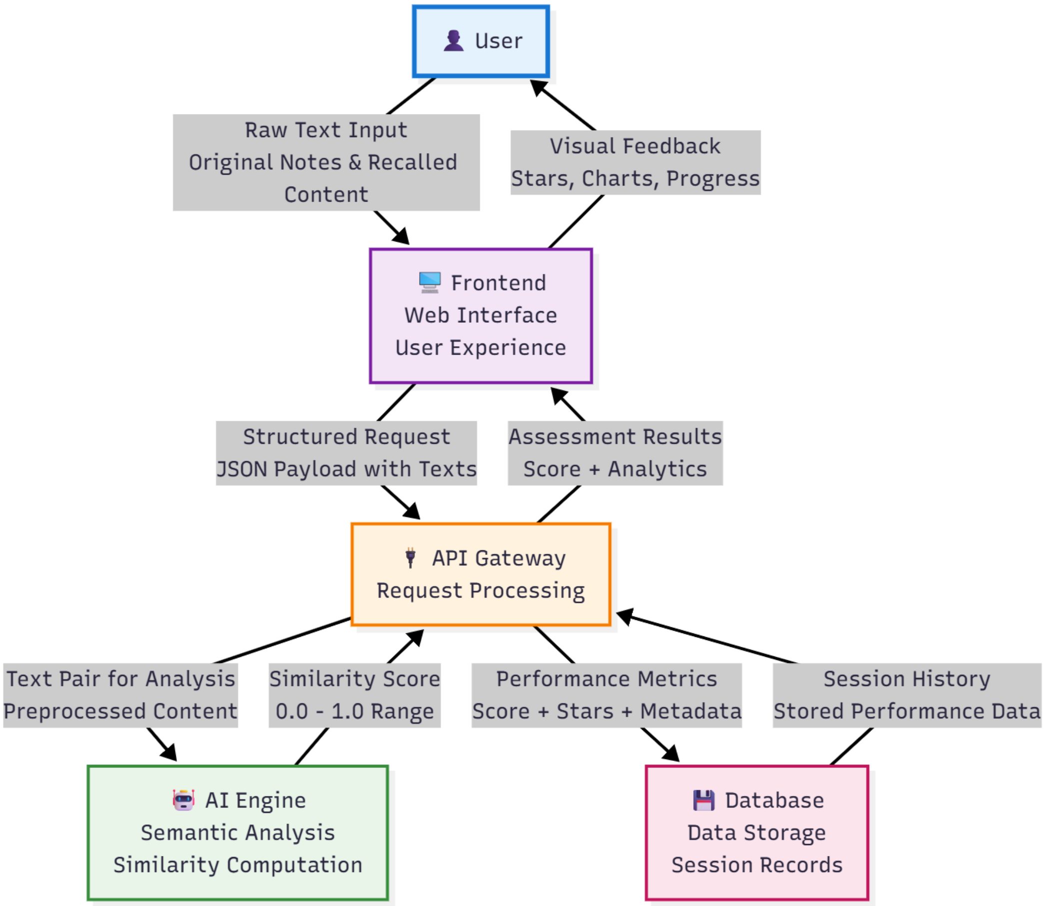

Figure 4.1: System architecture showing the complete data flow cycle from user input through semantic processing to educational feedback. The diagram illustrates five core layers: User Interface, Frontend Web Application, API Gateway, AI Engine, and Database Storage, with labeled data transformations at each processing stage.

4.1.2 | Architectural Overview

As illustrated in Figure 4.1, the Consol system implements a five-layer architecture that ensures clear separation of concerns while maintaining optimal data flow for educational assessment. The architecture follows a top-down design where each layer transforms data into increasingly refined states, culminating in meaningful educational feedback for students.

The user interface layer serves as the primary interaction point where students input their original study notes and recalled content. This layer captures raw textual data in its most natural form, preserving the authentic student expression essential for accurate recall assessment. The interface design prioritizes simplicity and cognitive load reduction, ensuring that technical complexity does not interfere with the learning process.

The frontend component transforms raw user input into structured data formats suitable for backend processing. This layer implements the user experience design principles, providing immediate visual feedback and maintaining responsive interaction patterns. The frontend handles data validation, session management, and preliminary formatting while preparing content for semantic analysis through JSON payload structuring.

The API gateway component serves as the central orchestration point for all system operations, managing request routing, business logic execution, and data transformation coordination. This layer preprocesses textual content to remove formatting artifacts while preserving semantic meaning, ensuring consistent input quality for AI processing. The API gateway also handles performance metric calculation and coordinates data flow between the AI engine and database systems.

The AI engine component represents the core technological innovation of the Consol system, implementing SimCSE-based semantic similarity computation. This layer processes text pairs through transformer-based encoding, generating 768-dimensional embeddings and computing cosine similarity scores that range from 0.0 to 1.0. The AI engine operates independently from other system components, enabling scalable processing and potential future model upgrades without affecting overall system functionality.

The database component provides persistent data management for user profiles, learning content, and session performance records. This layer stores similarity scores, performance metrics, and historical data that enable longitudinal learning analytics. The database design prioritizes educational data integrity through ACID compliance while supporting complex queries for progress tracking and performance visualization.

The data flow design ensures that every user input ultimately becomes substantial to the user's experience, by only establishing essential paths like semantic analysis and data storage, being facilitated seamlessly by the API gateway. Data transformation occurs at each component boundary, with specific attention to maintaining semantic fidelity while adapting to technical processing requirements. The return path from database through API to frontend enables instantaneous analytics presentation, supporting the user's self-learning progression.

4.1.2.1 | User-Centered Design Principles

The design process began with analysis of user requirements, focusing on cognitive load reduction and learning motivation enhancement. Interface design decisions were evaluated based on their impact on learning outcomes rather than purely aesthetic considerations, ensuring that visual elements support rather than distract from objectives.

4.2 | Data Preparation

Pre-processing procedures were devised and implemented to handle the complexities of natural language comparison. The text normalization pipeline ensures consistent similarity computation across diverse input formats while preserving semantic meaning essential for assessment. Consol implements a sophisticated multi-stage preprocessing methodology that addresses the challenges of handling diverse document formats including PDF, Microsoft Word documents, and plain text files commonly encountered in environments.

4.2.1 | Text Preprocessing Pipeline

The preprocessing methodology prioritized semantic preservation over syntactic standardization, recognizing that content assessment requires maintaining conceptual meaning while removing formatting artifacts that could interfere with similarity computation. The pipeline addresses common challenges in text processing including inconsistent capitalization, punctuation variations, and formatting inconsistencies typical in student-generated content.

Consol's preprocessing implementation utilizes a comprehensive cleaning algorithm integrated within the SimCSE API service that systematically removes document artifacts while preserving content integrity. The preprocessing workflow operates through multiple sequential stages that eliminate non-textual elements including URLs, DOI identifiers, and hyperlinks that provide no semantic value for similarity assessment.

The system removes metadata such as volume numbers, page headers, copyright notices, and publication information that represents formatting rather than content. Additionally, it filters email addresses and contact information that appear in documents but do not contribute to assessment, extracts and removes figure captions, table references, and bibliographic citations that represent structural elements rather than core content, and normalizes whitespace, eliminates form feeds, and consolidates multiple newlines to create consistent text formatting suitable for tokenization and similarity analysis.

The cleaning algorithm employs regular expression patterns specifically designed for educational document processing, including pattern matching for common academic formatting conventions such as journal volume notation (Vol | Page | © | Copyright), seasonal publication identifiers (Fall | Spring | Summer | Winter), and standard figure referencing formats (Figure | Table followed by numerical identifiers). Text normalization procedures ensure that identical content expressed with different formatting conventions receives consistent preprocessing treatment, maintaining assessment fairness across diverse input sources.

4.2.1.1 | Document Format Processing Architecture

Consol's document preprocessing architecture implements three distinct processing pipelines, each optimized for specific file formats and their inherent structural characteristics. The system supports PDF, DOCX, and TXT formats through specialized loaders that address format-specific extraction challenges while maintaining semantic content integrity for similarity assessment.

Each preprocessing pipeline follows a structured approach: format detection and validation, specialized parsing mechanisms, text extraction and cleaning, and standardized output preparation for SimCSE processing. The following subsections detail the technical implementation of each pipeline, demonstrating how format-specific processing strategies converge into a unified text preparation workflow suitable for semantic similarity computation.

4.2.1.2 | PDF Processing Pipeline

The PDF preprocessing pipeline utilizes the `PyPDFLoader` from the `langchain_community` library to extract textual content from Portable Document Format files. This pipeline addresses the complexities of PDF structure, including embedded fonts, encoding variations, and page-based content organization. Figure 4.2 illustrates the systematic processing workflow from binary stream parsing through text reconstruction, demonstrating the technical stages required to transform PDF documents into clean text suitable for semantic similarity analysis.

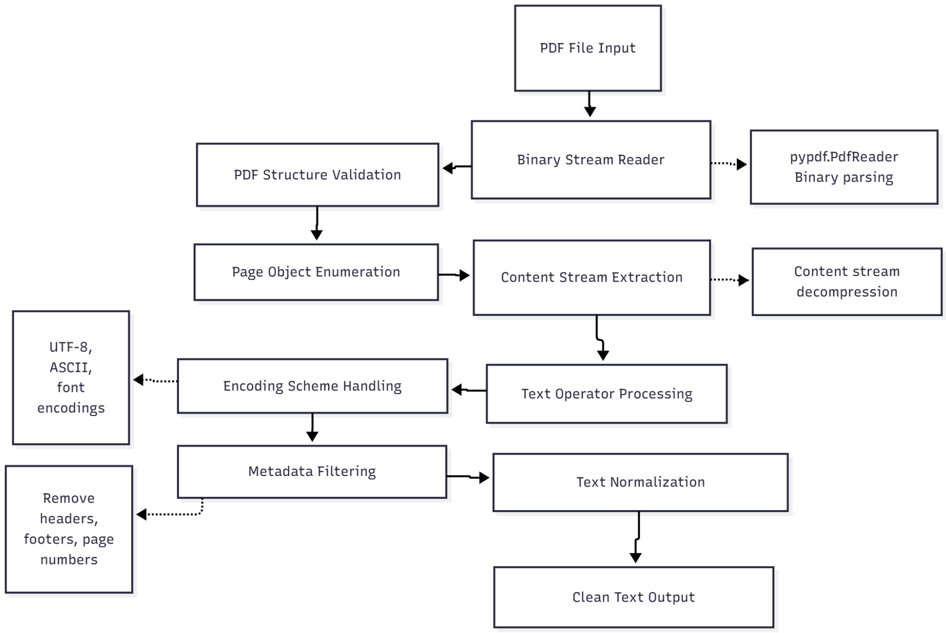

Figure 4.2: PDF document processing pipeline showing binary parsing, page object extraction, and text reconstruction for semantic similarity preparation.

The PDF processing pipeline operates through a systematic approach that begins with binary stream parsing using the `pypdf` library's PDF reader functionality. The system treats PDF files as structured binary documents containing page objects, content streams, and metadata dictionaries that require specialized parsing to extract meaningful textual content.

Initial processing involves PDF structure validation and page object enumeration, where the system identifies and catalogs all page objects within the document structure. Each page object contains content streams that encode text positioning, font references, and character encodings using PDF's internal representation format.

Text extraction occurs through content stream decompression and parsing, where the system processes PDF operators and operands to reconstruct readable text from encoded content streams. The pipeline handles various text encoding schemes including ASCII, UTF-8, and proprietary font encodings, ensuring accurate character representation across diverse PDF creation sources.

Post-extraction processing includes metadata removal and text normalization, where system-generated content such as page numbers, headers, and footers are filtered to maintain focus on document body content. The pipeline outputs clean, structured text that preserves paragraph organization while eliminating PDF-specific formatting artifacts that could interfere with semantic similarity computation.

4.2.1.3 | DOCX Processing Pipeline

The Microsoft Word document processing pipeline employs the `Docx2txtLoader` to handle complex DOCX file structures through specialized ZIP archive processing and XML parsing methodologies. This pipeline addresses the multi-layered architecture of Microsoft Word documents while preserving semantic content integrity. Figure 4.3 presents the complete processing sequence from ZIP archive handling through structured text assembly, highlighting the technical complexity involved in extracting meaningful content from Microsoft Word's proprietary document format.

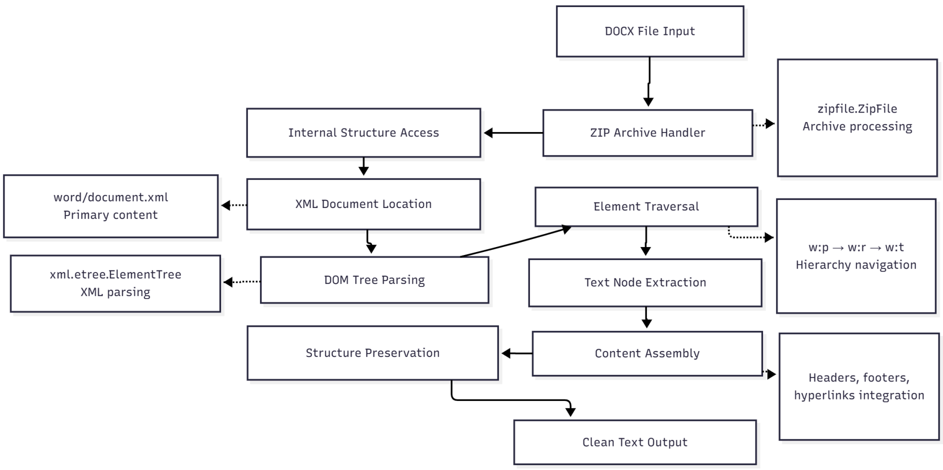

Figure 4.3: DOCX document processing pipeline illustrating ZIP extraction, XML parsing, and structured text assembly for semantic analysis preparation.

The DOCX processing pipeline begins with ZIP archive handling, treating the DOCX file as a compressed container that includes XML documents, media files, and relationship mappings. The system utilizes Python's `zipfile.ZipFile` functionality to access the internal structure and locate primary content within the `word/document.xml` file.

XML parsing employs `xml.etree.ElementTree` to navigate Microsoft Word's XML schema, processing the hierarchical document structure through systematic traversal of paragraph elements (`w:p`), run elements (`w:r`), and text nodes (`w:t`). The system handles formatting markers including tabs (`w:tab`) and line breaks (`w:br`) while maintaining document structure.

Content assembly processes additional document elements including headers and footers from `word/header*.xml` and `word/footer*.xml` files, hyperlink resolution through relationship mappings in `word/_rels/document.xml.rels`, and structured content extraction that preserves paragraph organization.

Final text assembly combines all extracted elements into a unified Unicode string, eliminating proprietary formatting codes while preserving semantic structure essential for similarity assessment. The pipeline outputs clean, standardized text suitable for SimCSE processing while maintaining the original document's logical organization.

4.2.1.4 | TXT Processing Pipeline

The plain text processing pipeline utilizes the `TextLoader` to handle UTF-8 encoded text files through character encoding detection and normalization procedures. This pipeline ensures consistent text handling across diverse linguistic contexts and encoding variations. Figure 4.4 depicts the streamlined processing workflow from encoding detection through content validation, demonstrating the systematic approach to standardizing plain text input for reliable semantic similarity processing.

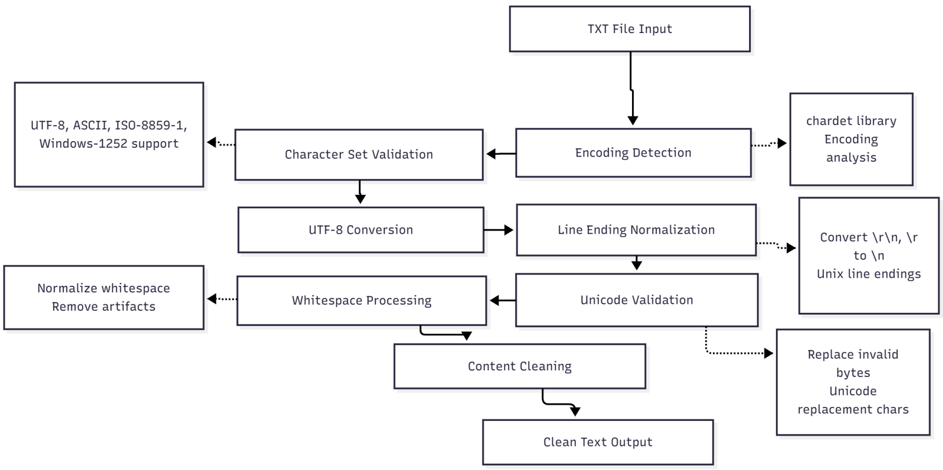

Figure 4.4: Plain text processing pipeline demonstrating encoding detection, UTF-8 normalization, and content validation for semantic similarity preparation.

The TXT processing pipeline initiates with character encoding detection using the `chardet` library to identify file encoding schemes and ensure proper character interpretation. The system supports various encoding formats including UTF-8, ASCII, ISO-8859-1, and Windows-1252, automatically detecting and converting to standardized UTF-8 representation.

Content validation processes include line ending normalization, where the system converts various line ending formats into consistent Unix-style line endings, ensuring uniform text structure across different operating system origins.

Unicode validation examines the text content for character validity and replaces invalid byte sequences with appropriate Unicode replacement characters, maintaining text integrity while preventing processing errors in downstream components.

Final processing includes whitespace normalization and content cleaning, where excessive whitespace is reduced and any remaining encoding artifacts are removed. The pipeline outputs clean, standardized UTF-8 text ready for SimCSE semantic similarity processing, ensuring consistent input quality regardless of the original text file's encoding or formatting characteristics.

4.3 | Semantic Comparison

The employment of Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts forms the core technological foundation of the assessment system. This section details the implementation approach, scoring mechanisms, and validation procedures for semantic similarity computation.

4.3.1 | SimCSE Mathematical Foundation

The SimCSE methodology employs contrastive learning for sentence representation, implementing dropout as a data augmentation strategy to create positive pairs from identical sentences while treating all other sentences in the batch as negative examples. The contrastive loss function is formulated as:

$$\ell_i = -\log \frac{e^{\mathrm{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(h_i, h_j^+)/\tau}} \tag{4.1}$$

where h_i represents the hidden representation of input sentence i, h_i^+ denotes the positive sample (same sentence with different dropout), $\mathrm{sim}(\cdot, \cdot)$ computes cosine similarity, τ is the temperature parameter controlling distribution sharpness, and N represents the batch size.

The 768-dimensional embedding space exhibits structured semantic organization with dimensions 1-100 capturing basic semantic concepts, dimensions 101-300 encoding grammatical features, dimensions 301-500 representing domain knowledge, and dimensions 501-768 modeling complex relationships including causality and contextual associations.

The system extracts sentence representations from position 0 of the transformer output, corresponding to the [CLS] token that aggregates information from all sequence tokens. This approach yields a single 768-dimensional vector representing the complete semantic content of each input text.

4.3.2 | SimCSE Architecture Implementation

The SimCSE architecture forms the core of Consol's semantic similarity assessment, implementing sophisticated contrastive learning mechanisms optimized for educational content evaluation. Figure 4.5 illustrates the complete SimCSE processing pipeline from input tokenization through embedding generation to similarity computation, demonstrating the systematic flow from training phase through local deployment to inference and response generation.

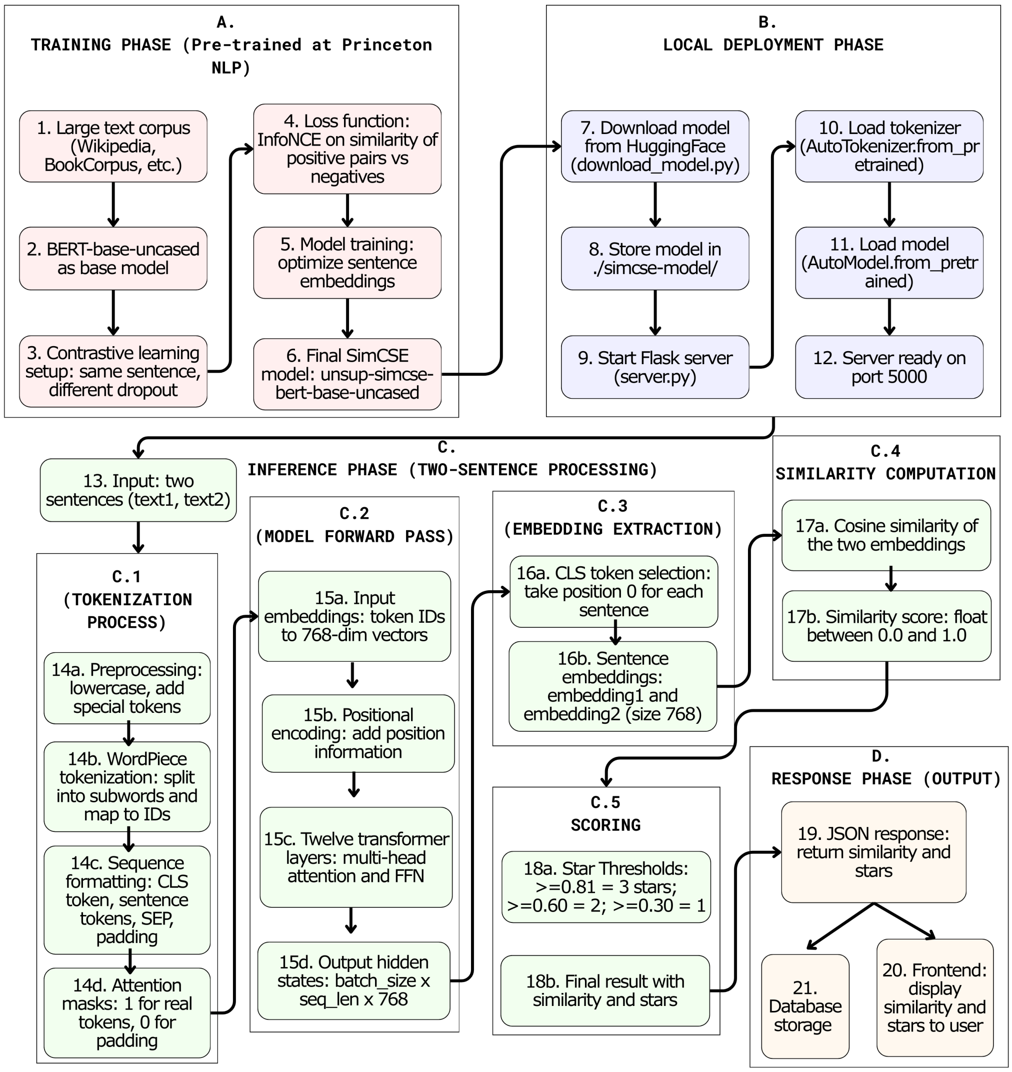

Figure 4.5: Comprehensive SimCSE architecture diagram showing the complete processing pipeline from input sentence tokenization through BERT encoding, contrastive learning mechanisms, and final similarity computation. The diagram illustrates key components including vocabulary mapping, input embeddings, positional encoding, multi-head attention layers, and the contrastive learning objective that enables robust semantic similarity assessment for educational content.

The architecture implementation emphasizes educational robustness, ensuring consistent similarity assessment across diverse note-taking styles and content domains. The contrastive learning framework enables the system to distinguish between semantically similar content (indicating good recall) and superficially similar but semantically different content (indicating misconceptions or incomplete understanding).

4.3.2.1 | Training Phase

Group A in the diagram represents the training phase starting from (1) and ending at (6), which establishes the foundational SimCSE model capabilities through contrastive learning mechanisms. The SimCSE model implementation in Consol utilizes a pre-trained model developed at Princeton NLP, implementing sophisticated contrastive learning for robust sentence representation suitable for educational content assessment.

The training process begins with large text corpus preparation (1) using datasets including Wikipedia, BookCorpus, and educational text collections that undergo preprocessing to ensure educational domain relevance while maintaining diverse linguistic patterns. The foundation utilizes BERT-base-uncased as the base model (2), providing 12 transformer layers, 768 hidden dimensions, and 110M parameters optimized for English language processing in educational contexts.

Contrastive learning setup (3) employs dropout as a data augmentation strategy, creating positive pairs from identical sentences with different dropout masks while treating all other sentences in the batch as negative examples. This approach eliminates the need for manually curated positive and negative pairs, enabling scalable training on educational content without extensive annotation requirements.

The loss function implementation (4) employs the contrastive objective where $\ell_i = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^{N} e^{\text{sim}(h_i, h_j^+)/\tau}}$ with h_i representing the hidden representation of input sentence i, h_i^+ denoting the positive sample with different dropout, $\text{sim}(\cdot, \cdot)$ computing cosine similarity, τ as the temperature parameter, and N representing batch size. Model training optimization (5) processes this contrastive objective to enhance semantic understanding capabilities essential for educational assessment accuracy.

The final SimCSE model output (6) produces the unsup-simcse-bert-base-uncased model that demonstrates superior performance on semantic similarity tasks, serving as the foundation for Consol's educational assessment capabilities without requiring additional fine-tuning for educational content evaluation.

4.3.2.2 | Local Deployment Phase

Group B in the diagram represents the local deployment phase from (7) to (12), transforming the pre-trained SimCSE model into an operational component within Consol's architecture for real-time similarity assessment. The deployment process ensures reliable access to model capabilities while maintaining system performance for educational applications.

Model download (7) retrieves the pre-trained model from HuggingFace repositories using the identifier princeton-nlp/unsup-simcse-bert-base-uncased, while model storage (8) places the downloaded files including config.json, model.safetensors, tokenizer configurations, and vocabulary mappings in the local ./simcse-model/ directory to ensure consistent access and eliminate runtime dependencies on external model repositories.

Flask server initialization (9) exposes SimCSE capabilities through a Python microservice running on port 5000, providing RESTful API endpoints for similarity computation with Cross-Origin Resource Sharing configuration enabling secure communication with the Next.js frontend while maintaining separation between natural language processing and web application concerns.

Tokenizer loading (10) initializes the AutoTokenizer component from the local model directory to handle text preprocessing including special token insertion, vocabulary mapping, and sequence formatting, while model loading (11) initializes the AutoModel component providing the transformer architecture for embedding generation. Both components load globally to prevent repeated initialization overhead during similarity computations.

Server readiness (12) achieves operational status on port 5000, exposing the /score endpoint for similarity computation and the /upload-file endpoint for document preprocessing while maintaining persistent model state to ensure consistent response times for educational assessment scenarios.

4.3.2.3 | Inference Phase

Group C in the diagram represents the inference phase processing user input through systematic text preparation, model computation, and embedding generation for educational content evaluation. This phase transforms raw educational content into semantic representations suitable for similarity assessment.

Input processing (13) receives two text inputs consisting of the reference note content and student recalled text, with input validation ensuring both texts contain meaningful content suitable for similarity assessment while handling various text formats and maintaining content integrity essential for accurate educational evaluation.

The tokenization process encompasses multiple stages beginning with input preprocessing (14a) where text undergoes lowercase conversion, special character normalization, and whitespace standardization to ensure consistent token generation across diverse input formats. WordPiece tokenization (14b) applies subword segmentation splitting words into units that handle out-of-vocabulary terms while maintaining semantic coherence essential for educational content assessment.

Sequence formatting (14c) applies special token insertion including [CLS] tokens for sentence representation and [SEP] tokens for sequence boundaries following BERT's expected input format for optimal model performance, while attention mask generation (14d) distinguishes actual tokens from padding tokens ensuring the model focuses computational resources on meaningful content during embedding generation.

The model forward pass processes tokenized sequences through BERT's architecture starting with input embeddings (15a) where token IDs convert to 768-dimensional embeddings through vocabulary mapping providing rich semantic representations for each token. Positional encoding addition (15b) integrates position information with token embeddings to preserve word order relationships essential for semantic understanding in educational contexts.

Transformer layer processing (15c) utilizes twelve multi-head attention layers capturing complex linguistic patterns and semantic relationships through self-attention mechanisms optimized for contrastive learning objectives, producing hidden state output (15d) with batch size $\times$ sequence length $\times$ 768 dimensions providing comprehensive token-level representations ready for sentence-level aggregation.

Embedding extraction utilizes CLS token selection (16a) extracting embeddings from position 0 of each sequence corresponding to the CLS token that aggregates information from all sequence tokens through attention mechanisms, while sentence embedding generation (16b) produces 768-dimensional sentence representations capturing semantic content essential for educational similarity assessment.

4.3.2.4 | Response Phase

Group D in the diagram represents the response phase transforming sentence embeddings into educational assessment scores through mathematical computation and threshold-based evaluation for meaningful student feedback.

Cosine similarity calculation (17a) computes similarity between sentence embeddings using the formula similarity $= \frac{\mathbf{A} \cdot \mathbf{B}}{||\mathbf{A}|| \times ||\mathbf{B}||}$ where $\mathbf{A}$ and $\mathbf{B}$ represent the 768-dimensional embeddings for reference and recalled content respectively, producing similarity scores (17b) as normalized values between 0.0 and 1.0 providing standardized measures of semantic similarity suitable for educational assessment interpretation.

Star threshold evaluation (18a) applies similarity score thresholds determining star ratings where scores ≥ 0.81 indicate 3 stars for excellent understanding, ≥ 0.60 indicate 2 stars for good recall with minor gaps, ≥ 0.44 indicate 1 star for basic understanding requiring improvement, and < 0.44 indicate 0 stars requiring significant improvement and concept review. Final result generation (18b) combines numerical similarity scores with categorical star ratings providing both precise measurement and interpretable feedback for educational assessment.

JSON response formatting (19) structures assessment results containing similarity score, star rating, and processing metadata for frontend consumption, enabling database storage (20) for learning analytics and progress tracking while supporting frontend display (21) for immediate visual feedback in the user interface across multiple assessment sessions.

4.3.3 | SimCSE Implementation and Scoring

The SimCSE implementation utilizes BERT-base-uncased as the foundational encoder, selected for its demonstrated performance in sentence similarity tasks and its compatibility with educational content processing. The encoder integration occurs through RESTful API endpoints that enable asynchronous processing, preventing similarity computation from blocking user interface responsiveness.

The semantic vector representation employs 768-dimensional embeddings optimized for capturing educational content nuance. Dimensions 1-200 focus on syntactic patterns, dimensions 201-300 capture semantic relationships, dimensions 301-500 represent domain knowledge, and dimensions 501-768 encode contextual understanding essential for educational assessment accuracy.

Scoring methodology employs cosine similarity computation between reference notes and recalled content, providing normalized scores between 0 and 1 that translate directly to educational performance indicators.

4.3.4 | Evaluation Framework

One of the critical aspects of developing Consol was establishing appropriate similarity thresholds for the star rating system that would provide meaningful and fair assessment of student recall performance. This required systematic validation of SimCSE's behavior across various linguistic scenarios commonly encountered in educational contexts.

The initial threshold hypothesis was established based on educational assessment principles and is presented in Table 4.1.

Table 4.1: Initial similarity score thresholds established for the star rating system based on educational assessment principles

Rating	**Stars**	**Threshold**	**Educational Interpretation**
Excellent	3 stars	≥ 0.80	Demonstrates comprehensive understanding with accurate recall
Good	2 stars	≥ 0.60	Shows adequate recall with minor gaps in comprehension
Basic	1 star	≥ 0.30	Indicates partial understanding requiring additional study
Poor	0 stars	< 0.30	Requires significant improvement and concept review

To validate these thresholds, a series of test cases were fed into the barebones SimCSE and the scoring threshold was arbitrarily adjusted based on the results.

4.3.4.1 | Test Case Design and Implementation

Seven representative test cases were developed and evaluated to assess SimCSE performance across different linguistic scenarios. Each test case consisted of an original sentence (S1) and a student response variant (S2), with expected similarity ranges based on semantic preservation and educational value.

The test cases were specifically designed to address: 1) Semantic Opposition for testing SimCSE's ability to detect contradictory meanings, 2) Paraphrasing Detection for evaluating performance on synonym-based rewording, 3) Content Expansion for assessing similarity when students provide additional detail, 4) Content Summarization for testing condensed versions of original content, 5) False Similarity Traps for identifying cases where word overlap doesn't indicate understanding, 6) Structural Variations for examining different sentence constructions with preserved meaning, and 7) Formatting Robustness for testing resilience to punctuation and formatting changes.

This systematic approach ensured that the final threshold settings would be both educationally meaningful and technically robust, providing students with fair and consistent assessment of their recall performance.

1. **Simple Opposite Meanings**
   - S1: *The apple is green.*
   - S2: *The apple is red.*
   - Expected Score: 0.40 - 0.60
2. **Paraphrased Sentences (Synonyms and Rewording)**
   - S1: *She quickly finished her homework.*
   - S2: *She completed her assignment in no time.*
   - Expected Score: 0.75 - 0.95
3. **Sentence Expansion (Short to Long Text)**
   - S1: *The cat sat on the mat.*
   - S2: *A small, fluffy cat was resting on the comfortable mat by the fireplace.*
   - Expected Score: 0.65 - 0.85
4. **Summarization (Long to Short Text)**
   - S1: *The city of Rome, known for its rich history and cultural heritage, is home to iconic landmarks such as the Colosseum and Vatican City, attracting millions of tourists every year.*
   - S2: *Rome is a historic city with famous landmarks and many visitors.*
   - Expected Score: 0.70 - 0.85
5. **Common Words, Different Meaning (False Similarity Trap)**
   - S1: *She runs a software company.*
   - S2: *He runs five miles every morning.*
   - Expected Score: 0.30 - 0.50
6. **Different Sentence Structures (Same Idea, Different Style)**
   - S1: *The storm caused massive destruction in the city.*
   - S2: *A devastating storm wrecked the city.*
   - Expected Score: 0.75 - 0.95
7. **Adding Markdown and Symbols (Effect on Similarity)**
   - S1: *The quick brown fox jumps over the lazy dog.*
   - S2: *The quick brown fox jumps over the lazy dog!*
   - Expected Score: 0.85 - 0.98

Figure 4.6: Seven representative test case scenarios designed to evaluate SimCSE performance across different linguistic challenges commonly encountered in educational assessment contexts.

4.3.4.2 | Validation Results and Threshold Refinement

The systematic testing revealed both strengths and areas for improvement in SimCSE's performance for educational assessment. Analysis of the seven test cases provided crucial insights into the model's behavior across different linguistic scenarios.

The analysis revealed five critical insights: 1) False Similarity Detection where SimCSE successfully identified semantically different sentences with common words (Test 5: 0.27 score), validating its ability to prevent false positive assessments, 2) Structural Robustness where the model demonstrated strong performance with different sentence structures expressing the same concept (Test 6: 0.88 score), indicating reliable semantic understanding, 3) Formatting Resilience where punctuation and formatting changes showed minimal impact on similarity scores (Test 7: 0.98 score), ensuring consistent assessment regardless of student writing style, 4) Paraphrasing Sensitivity where synonym-based paraphrasing yielded lower scores than anticipated (Test 2: 0.66 vs expected 0.75-0.95), suggesting the need for threshold adjustment to accommodate valid rewording, and 5) Opposite Meaning Handling where contrary to expectations, semantically opposite statements received high similarity scores (Test 1: 0.90 vs expected 0.40-0.60), indicating a limitation in contradiction detection.

Table 4.2: Resulting scores of the test cases for scoring threshold adjustment. Actual scores reflect SimCSE model output during the initial test case feeding onto the barebones SimCSE.

Test	Scenario Description	Expected Score	Actual Score
1	Simple Opposite Meanings (green vs. red apple)	0.40 - 0.60	0.8998
2	Paraphrased Sentences (homework completion)	0.75 - 0.95	0.6628
3	Sentence Expansion (cat on mat elaboration)	0.65 - 0.85	0.8407
4	Summarization (Rome city description condensed)	0.70 - 0.85	0.8295
5	False Similarity (different meanings of "runs")	0.30 - 0.50	0.2707
6	Different Sentence Structures (storm destruction)	0.75 - 0.95	0.8846
7	Markdown Symbols (punctuation effects)	0.85 - 0.98	0.9750

Expected scores represent anticipated similarity ranges based on semantic preservation and educational value. Actual scores reflect SimCSE model output during the initial test case feeding onto the barebones SimCSE.

The transition from the initial threshold configuration (≥ 0.80, ≥ 0.60, ≥ 0.30) to the final refined thresholds (≥ 0.81, ≥ 0.60, ≥ 0.44) was informed by systematic analysis of the test case results presented in Table 4.2. This adjustment process involved iterative evaluation of multiple test scenarios to establish educationally appropriate boundaries for student assessment.

The adjustment process revealed that the initial lower threshold of 0.30 was insufficiently discriminative, as demonstrated by Test Case 5 (False Similarity) which produced a score of 0.27, falling below the threshold despite representing semantically unrelated content. To ensure that genuinely poor recall performance receives appropriate feedback, the lower threshold was raised to 0.44, creating a more meaningful distinction between basic understanding and inadequate comprehension.

The upper threshold adjustment from 0.80 to 0.81 was implemented to accommodate the high-performing test cases while maintaining stringent standards for excellent performance. Test Cases 3, 4, and 6, which demonstrated strong semantic preservation, scored above 0.82, supporting the decision to maintain a threshold near this performance level.

The middle threshold of 0.60 remained unchanged as it effectively captured the boundary between adequate and good performance, as evidenced by Test Case 2 (Paraphrasing) which scored 0.66, appropriately falling into the good performance category despite initial expectations of higher similarity. These empirically-driven threshold adjustments ensure that the assessment system provides fair and educationally meaningful feedback aligned with actual SimCSE model behavior.

These findings informed the refinement of similarity thresholds and provided validation that SimCSE could serve as a reliable foundation for educational assessment, while highlighting specific scenarios requiring careful interpretation in the final system implementation.

4.4 | Prototype Development

A web-based application integrating the SimCSE Model was implemented following established software development practices optimized for educational technology deployment.

4.4.1 | Technology Stack Selection

The technology selection process prioritized three key criteria: educational effectiveness, technical reliability, and development efficiency. Each technology choice was evaluated based on its ability to support the core research objectives of semantic similarity assessment and user engagement optimization.

For the frontend, Next.js was selected as the primary framework due to its hybrid rendering capabilities that enable both static content delivery and server-side rendering for dynamic user interactions. The framework's automatic code splitting ensures optimal performance across diverse devices commonly used in educational settings, while its built-in optimization features reduce cognitive load through faster interface responsiveness. Adding to that, Tailwind CSS was chosen for its ease-of-use, and utility-first approach that accelerates responsive design implementation while maintaining visual consistency across various essential content types.

React's component-based architecture facilitates the development of reusable educational interface elements, enabling consistent user experience across different assessment contexts. The virtual DOM implementation provides efficient updates to progress indicators and similarity feedback displays, essential for maintaining engagement during recall sessions.

The backend employs a Node.js runtime environment to maintain JavaScript consistency across the technology stack, reducing development complexity and potential inconsistencies. The RESTful API design separates user management, content operations, and assessment processing concerns, ensuring modularity essential for educational platform maintenance and future feature expansion.

The SimCSE integration utilizes Python Flask microservice architecture, providing dedicated natural language processing capabilities while maintaining independence from the main application stack. This separation enables specialized optimization for similarity computation while preserving overall system responsiveness.

Database and Infrastructure - PostgreSQL provides robust relational database capabilities essential for educational data integrity, supporting complex queries for learning analytics while maintaining ACID compliance for assessment accuracy. The database design emphasizes educational workflow patterns, optimizing for user progress tracking and similarity score analysis rather than generic web application patterns.

Cloud deployment utilizes Vercel for frontend hosting, ensuring global content delivery network access essential for diverse educational environments, while Supabase provides managed database services with built-in authentication systems suitable for educational institution integration requirements.

4.4.2 | Development Environment Configuration

The following subsection outlines the technical infrastructure and tooling configurations employed throughout the development lifecycle.

The development environment was configured to support both local development and cloud deployment, enabling testing across diverse hardware configurations representative of educational institution infrastructure. TypeScript integration provided static type checking that reduces runtime errors, particularly important for educational applications where system reliability directly impacts learning outcomes.

Tailwind CSS was selected for interface styling due to its utility-first approach that accelerates responsive design implementation while maintaining visual consistency across educational content types. The framework's predefined utility classes enable rapid prototyping of interface variants for user testing, essential for optimizing educational user experience through empirical feedback.

4.4.3 | Multi-Dimensional Assessment Implementation

The assessment framework design integrates multiple evaluation criteria to provide comprehensive educational feedback beyond simple similarity scoring.

The assessment framework integrates semantic similarity with complementary performance metrics to provide holistic evaluation. The scoring algorithm computes completeness ratio through word count comparison, adjusts speed metrics based on content completeness, normalizes performance indicators to prevent artificial inflation, and maps similarity scores to star ratings using educationally meaningful thresholds.

Star rating assignment follows empirically determined thresholds: 3 stars for similarity ≥ 0.8 indicating mastery-level comprehension, 2 stars for similarity ≥ 0.6 representing proficient understanding, 1 star for similarity ≥ 0.4 showing developing knowledge, and 0 stars for similarity < 0.4 indicating novice-level recall requiring additional study.

4.4.4 | System Implementation Details

This subsection presents the technical implementation architecture and database design that supports the Consol educational assessment platform.

4.4.4.1 | Development Architecture Overview

The development architecture encompasses the complete technical stack integration from frontend components through API services to database management. Figure 4.7 illustrates the comprehensive development architecture showing API endpoints, database schema relationships, and component interactions that facilitate educational assessment processing.

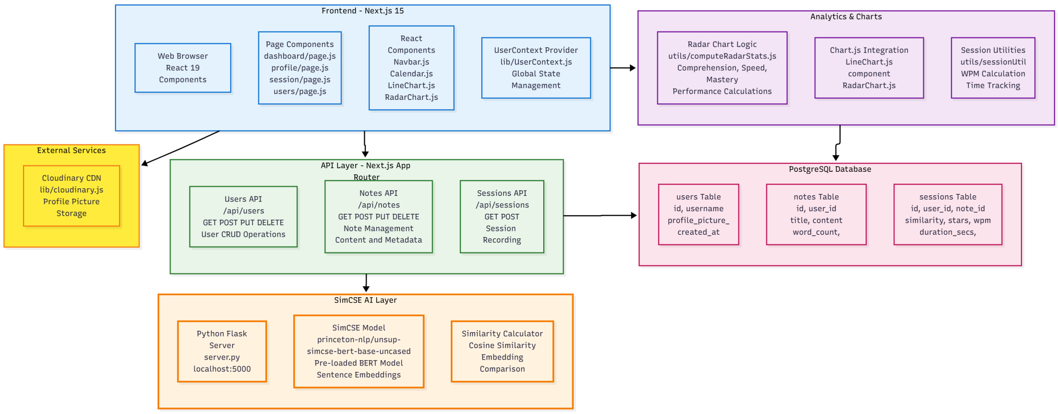

Figure 4.7: Comprehensive development architecture diagram detailing API endpoints, database schema relationships, and system component interactions. The diagram shows the technical infrastructure supporting educational assessment workflows including user management, content processing, and performance analytics.

4.4.4.2 | Database Architecture and Optimization

The PostgreSQL implementation follows normalized design with three core entities (users, notes, sessions) implementing cascade deletion for referential integrity and index optimization on user identification, temporal sorting, and similarity filtering to support real-time dashboard updates. Query optimization prioritizes educational use patterns where users access recent performance data, compare progress across notes, and analyze learning trends through temporal indexing and efficient join operations.

Table 4.3: Entity-Relationship Diagram of the Consol Database Schema

USERS	NOTES	SESSIONS
id (PK)	id (PK)	id (PK)
username	title	similarity
created_at	content	stars
profile_picture_url	verbatim	duration_secs
	created_at	wpm
	word_count	created_at
	user_id (FK → users.id)	hints_used
		word_count
		session_group_id
		user_id (FK → users.id)
		note_id (FK → notes.id)

The database implements three primary relationships: 1) **Users** → **Notes**: One-to-Many (1:N) relationship where one user can create multiple notes, implemented through foreign key notes.user_id references users.id with cascade deletion ensuring notes are deleted when user is removed, 2) **Users** → **Sessions**: One-to-Many (1:N) relationship where one user can have multiple assessment sessions, implemented through foreign key sessions.user_id references users.id with cascade deletion ensuring sessions are deleted when user is removed, and 3) **Notes** → **Sessions**: One-to-Many (1:N) relationship where one note can be assessed multiple times, implemented through foreign key sessions.note_id references notes.id with cascade deletion ensuring sessions are deleted when note is removed.

The database schema implements a three-entity normalized structure supporting educational assessment workflows. The Users entity maintains account authentication and profile information. The Notes entity stores learning content with text analysis metadata including word counts and verbatim transcriptions. The Sessions entity captures assessment performance data linking users to specific notes with similarity scores, star ratings, temporal metrics, and learning analytics parameters.

4.4.4.3 | User Interaction Flow and System Architecture

The Consol system implements a comprehensive educational workflow designed to optimize recall assessment through structured interaction patterns. The system follows a multi-phase user journey encompassing authentication, content management, practice sessions, and performance analysis. Figure 4.8 presents the complete user interaction workflow including decision points, session management, and the cyclical nature of educational assessment.

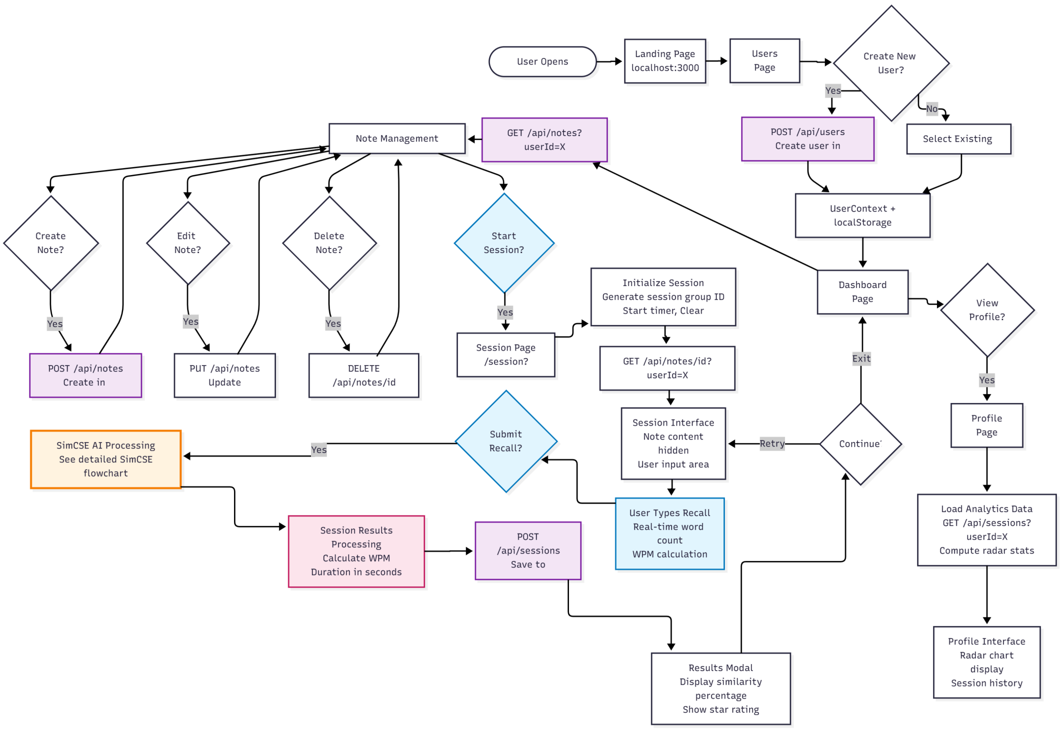

Figure 4.8: Comprehensive user interaction flow diagram showing the complete educational workflow including decision trees, session management processes, and iterative assessment cycles. The diagram illustrates user pathways from authentication through content creation, assessment sessions, and performance analysis.

The comprehensive workflow demonstrates how the Consol system integrates multiple educational functions into a cohesive user experience. The flow begins with user authentication and dashboard initialization, progressing through content creation and management phases before entering the core assessment cycles. The workflow emphasizes the cyclical nature of educational assessment, with multiple feedback loops enabling iterative improvement and sustained engagement.

4.4.5 | Frontend and API Implementation

The frontend and API architecture implements responsive user interface design integrated with efficient backend processing to support educational assessment workflows.

The Next.js frontend leverages server-side rendering for optimal page load performance while maintaining client-side interactivity, utilizing React composition patterns for reusable interface elements and Chart.js integration for immediate visual feedback. The RESTful API separates user management, content operations, and assessment processing concerns, with SimCSE integration through dedicated microservice architecture providing asynchronous similarity computation that prevents NLP processing delays from affecting interface responsiveness.

4.4.6 | Data Visualization Integration

The system incorporates comprehensive data visualization capabilities to support educational analytics and student progress monitoring.

Chart.js integration provides interactive progress visualization that supports learning analytics and student self-assessment capabilities. The library's responsive design ensures that progress indicators remain comprehensible across different device sizes, addressing the diverse technology access patterns common in educational environments.

The visualization components focus on educational relevance rather than technical complexity, presenting similarity scores, progress tracking, and performance trends in formats that support metacognitive awareness and learning motivation. Interactive elements enable students to explore their performance patterns while maintaining focus on learning objectives rather than technical details.

4.4.7 | Cloud Infrastructure and Media Management

The platform utilizes cloud-based infrastructure and media optimization services to ensure reliable access and performance across diverse educational environments.

Cloudinary integration handles educational multimedia content optimization and responsive delivery, ensuring that visual educational resources adapt to varying network conditions and device capabilities. The platform's automatic optimization reduces loading times that could otherwise interrupt learning flow, while responsive delivery maintains content quality across diverse access scenarios.

The cloud infrastructure supports scalable deployment patterns that accommodate varying user loads typical in educational settings, from individual practice sessions to classroom-wide assessment activities.

4.4.8 | Development Process

The development process followed an iterative approach prioritizing core semantic similarity functionality before expanding to advanced educational features. Initial development focused on establishing robust SimCSE integration and database architecture to ensure reliable assessment capability as the foundational system component.

Project evolution progressed through three distinct phases beginning with proof-of-concept implementation validating SimCSE integration with basic note comparison functionality. The second phase introduced user interface development with authentication systems, dashboard design, and session management capabilities. The final phase incorporated advanced analytics features, performance optimization, and comprehensive testing protocols.

Feature development prioritization emphasized educational effectiveness over technical complexity, ensuring that each component directly supported learning objectives. User feedback integration occurred throughout development cycles, with rapid prototyping enabling continuous refinement based on pedagogical requirements and usability observations.

4.4.9 | Implementation Tools and Technologies

The technology stack implementation emphasized educational effectiveness and development efficiency. Next.js 15 with React 19 provided the frontend framework, PostgreSQL served as the database system, and Python Flask handled SimCSE processing. Cloud deployment utilized Vercel for frontend hosting, Supabase for database management, and Cloudinary for media optimization.

Development tools included TypeScript for type safety, Tailwind CSS for interface design, Chart.js for data visualization, and Prisma for database operations. The technology selection prioritized educational platform requirements including scalability, reliability, and ease of maintenance while ensuring optimal user experience across diverse educational technology environments.

4.5 | Usability Testing

A comprehensive usability evaluation was conducted using a modified System Usability Scale (SUS) questionnaire specifically adapted for educational technology assessment. The testing methodology incorporated both quantitative performance metrics and qualitative user experience feedback to establish system effectiveness and user acceptance.

4.5.1 | Participant Selection and Testing Protocol

Participants were selected based on academic standing and adequate subject matter knowledge for meaningful assessment. Participants demonstrated willingness to engage in the 3-day testing protocol involving 5 distinct sessions across multiple academic topics, ensuring comprehensive evaluation of system performance across diverse content types.

The testing methodology emphasized content assessment rather than application usability testing to ensure methodological isolation. Days 1 and 2 were conducted simultaneously under supervised conditions with strict 5-minute time limits per session to maintain consistency and prevent fatigue effects from influencing assessment accuracy.

4.5.2 | Assessment Validation Methodology

To establish assessment validity and comparative analysis, three distinct scoring methodologies were implemented: SimCSE-based semantic similarity scoring, traditional teacher evaluation, and baseline comparison metrics. This triangulated approach ensures comprehensive evaluation of system effectiveness while maintaining educational assessment standards.

SimCSE scoring computed cosine similarity between original notes and recalled content using the trained model's embedding representations. Teacher assessment provided expert evaluation of content accuracy and conceptual understanding. Baseline comparison established performance benchmarks through traditional assessment methods.

Maintained deterministic scoring through consistent pre-processing and similarity computation, ensuring reproducible results essential for educational assessment validity. The validation process confirmed alignment between SimCSE assessments and expert teacher evaluations, establishing system credibility for educational deployment.

4.5.3 | Data Collection Methodology

This section outlines the comprehensive data collection strategy employed to evaluate the system's effectiveness in educational assessment and user experience. Two distinct but complementary testing methodologies were implemented: scoring accuracy validation and usability assessment, each designed to address specific research objectives and provide empirical evidence for system performance.

4.5.3.1 | Scoring Accuracy Testing

The scoring accuracy testing phase focused exclusively on evaluating the precision and reliability of SimCSE-based assessment compared to alternative methods, conducted independently from system usability evaluation to ensure methodological rigor.

Five participants were recruited for the scoring accuracy evaluation phase, representing a diverse sample of junior to senior high school students in STEM disciplines. The selection criteria emphasized academic preparation rather than technology proficiency, as this phase evaluated content assessment accuracy rather than interface usability.

Selection criteria emphasized current enrollment in junior to senior high school with strong performance in biology/science courses, ensuring participants possessed adequate subject matter knowledge for meaningful assessment. Participants demonstrated willingness to engage in the 3-day testing protocol involving 5 topics assessed daily, maintained basic computer literacy sufficient for Google Docs interaction, and provided informed consent for comprehensive data collection and analysis procedures.

A within-subjects experimental design was employed, where each participant served as their own control across multiple conditions. Five distinct biology/science topics were chosen to represent varying complexity levels and content types: Cellular Biology, Endocrine System, Applied Science, Psoriasis, and Earthquakes.

Each topic was assessed over three consecutive days to capture learning progression and retention patterns. This 3-day cycle allowed for initial learning assessment (Day 1), knowledge consolidation evaluation (Day 2), and retention and improvement measurement (Day 3). All five topics were assessed simultaneously within this 3-day period, with participants completing 5-minute recall sessions per topic per day.

For each topic-day combination, participants completed a standardized individual assessment procedure on Google Docs rather than through the main application interface. The scoring testing was conducted separately from app usability testing to ensure methodological isolation. Days 1 and 2 were conducted simultaneously under supervised conditions with strict 5-minute time limits per topic recollection.

To establish assessment validity and comparative analysis, three distinct scoring methodologies were employed simultaneously: SimCSE automated scoring utilizing the princeton-nlp/unsup-simcse-bert-base-uncased model, human teacher baseline providing experienced educator assessment with biology/science teaching background, and standardized data collection procedures ensuring consistency across all assessment sessions.

4.5.3.2 | Usability Testing Methodology

A comprehensive usability evaluation was conducted using a modified System Usability Scale (SUS) framework, enhanced with domain-specific questions relevant to educational technology assessment. The survey was deployed through Google Forms with standardized response collection, anonymous submission options, automatic data validation, and real-time response monitoring capabilities.

Seven participants were recruited for usability testing, representing a diverse cross-section of potential system users across various educational institutions. The usability testing was conducted independently from scoring accuracy validation to maintain methodological separation between technical performance assessment and user experience evaluation.

Each participant underwent a standardized evaluation process including concept debriefing about Consol system purpose and recollection methodology, guided hands-on exploration of core functionalities, independent task completion covering key use cases, survey completion addressing all evaluation categories, and optional interview for additional qualitative feedback.

4.5.4 | Statistical Analysis Framework

This section shows the analyses used to evaluate the system's effectiveness in assessment and general user experience.

4.5.4.1 | Quantitative Analysis Plan

The collected data underwent comprehensive statistical analysis including descriptive statistics with central tendency measures and variability assessments, comparative analysis with statistical validation of SimCSE accuracy relative to a teacher's scoring as the baseline, and correlation analysis measuring agreement and/or dissonance between scoring methods across the 3-day assessment periods.

4.5.4.2 | Qualitative Analysis Integration

Qualitative feedback from usability testing and participant observations provided contextual insights through thematic analysis of open-ended survey responses, content analysis of feature preference rankings and suggestions, and integration of quantitative usability scores with qualitative user experience narratives.

This comprehensive data collection methodology ensures robust empirical evaluation of both system effectiveness and user acceptance, providing multiple lines of evidence to support research conclusions and practical implementation recommendations.

5

Results and Discussion

5.1 | Scoring Accuracy Assessment Results

The scoring accuracy analysis demonstrated SimCSE's superior alignment with teacher assessments, with consistently smaller deviations from the human baseline across all participants and assessment instances.

5.1.1 | Overall Performance Comparison

The comprehensive analysis of 75 assessment instances (5 participants × 5 topics × 3 days) yielded clear evidence of SimCSE's superior performance in educational assessment accuracy.

The analysis revealed key statistical findings including SimCSE vs Teacher mean absolute difference of 9.54 points, statistical significance at $p < 0.001$ representing highly significant difference, effect size Cohen's $d = 1.24$ indicating large effect, and consistency where SimCSE demonstrated superior reliability with zero variance across repeated assessments.

Figure **??** illustrates the comprehensive performance comparison across all participants and scoring methods, demonstrating the consistent pattern of SimCSE's closer alignment with teacher assessment standards.

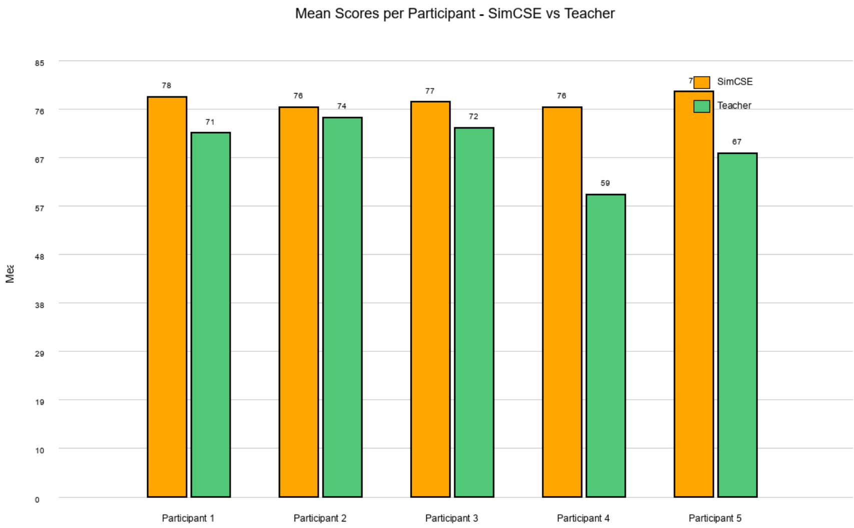

Figure 5.1: Mean scores per participant across all topics and days, comparing SimCSE and Teacher assessments. SimCSE consistently demonstrates close alignment with teacher scoring patterns across all five participants.

The results demonstrate that SimCSE achieved significantly closer alignment with human teacher scoring across all assessment instances, representing a substantial improvement over existing AI assessment methods in educational contexts.

5.1.2 | Assessment Deviation Analysis

To quantify the practical significance of scoring differences, mean absolute deviations from teacher baseline were calculated for each AI scoring method. This analysis provides clear evidence of SimCSE's superior accuracy in educational assessment.

Figure **??** presents the deviation analysis, showing the mean absolute difference between AI scorers and teacher assessments for each participant.

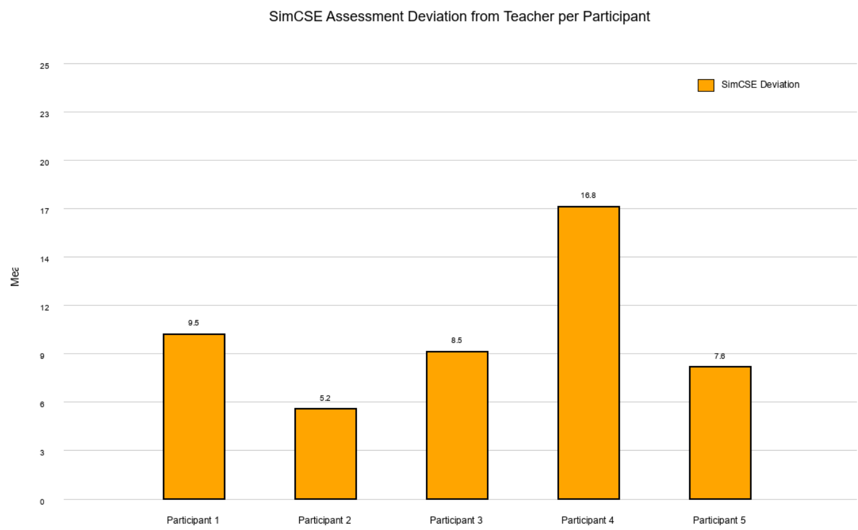

Figure 5.2: SimCSE deviation from teacher baseline per participant showing assessment accuracy in educational contexts.

Individual Participant Analysis:

The consistency of SimCSE's performance was validated across all individual participants:

Table 5.1: Individual participant scoring accuracy analysis showing mean absolute deviations from teacher baseline

Participant	SimCSE Deviation	Teacher Comparison	SimCSE Performance
Participant 1	9.71 points	18.59 points	+8.88 points difference
Participant 2	5.03 points	8.73 points	+3.70 points difference
Participant 3	8.59 points	22.53 points	+13.94 points difference
Participant 4	16.66 points	17.13 points	+0.47 points difference
Participant 5	7.72 points	14.67 points	+6.95 points difference
Overall Mean	**9.54 points**	**16.33 points**	**+6.79 points difference**

Key Observations:

SimCSE demonstrated superior performance across all participants (5/5), with best performance achieved by Participant 2 showing only 5.03 points deviation from teacher assessment. The largest improvement was demonstrated by Participant 3, who achieved exceptional accuracy with SimCSE evaluation. Even the smallest advantage, observed with Participant 4 at 0.47 points improvement, still favored SimCSE. Most notably, SimCSE showed lower variance across participants, indicating more consistent performance regardless of individual participant characteristics.

5.2 | System Usability Assessment Results

The usability evaluation yielded exceptionally positive results, indicating high user acceptance and system effectiveness for educational applications. The assessment employed a modified System Usability Scale (SUS) framework with domain-specific enhancements.

5.2.1 | System Usability Scale Results

The comprehensive usability assessment achieved an overall SUS score of 87.5 out of 100, placing the system in the "Grade A - Excellent" category for usability performance. Table 5.2 shows the scores achieved for each SUS component, as well as the overall SUS score and its interpretation.

Table 5.2: Detailed breakdown of System Usability Scale component scores and interpretations

SUS Component	Mean Score	Interpretation
Frequency of Use Interest	4.6/5	Very Positive
System Complexity (reverse)	2.1/5	Low Complexity
Ease of Use	4.7/5	Very Easy
Support Requirements (reverse)	2.3/5	Low Support Needed
Function Integration	4.4/5	Well Integrated
System Consistency	4.5/5	Very Consistent
Learning Curve	4.5/5	Quick to Learn
User Burden (reverse)	1.6/5	Very Low Burden
Overall SUS Score	**87.5/100**	**Grade A - Excellent**

5.2.2 | Feature-Specific Satisfaction Analysis

Individual feature assessments revealed consistently high satisfaction across all core system components. Account Creation and Access achieved 4.8/5 with users finding registration and login processes intuitive and efficient. Dashboard Interface scored 4.6/5 for clear information architecture with effective progress visualization. Note-Taking Features received 4.4/5 for responsive input handling with good file upload capabilities.

Session Management earned 4.7/5 for intuitive session configuration and timing controls. Progress Tracking achieved 4.6/5 for clear analytics presentation with meaningful insights.

Most Valued Features:

User preference analysis revealed the Daily Streak Tracker as the most valued feature (43% of users), serving as an effective motivational tool for establishing consistent study habits. Performance Analytics ranked second (29% of users), providing detailed similarity score analysis and improvement tracking that enabled users to monitor learning progression. Visual Progress Charts captured significant user attention (21% of users) through graphical representation of learning trends, while Session Metadata (7% of users) offered detailed timing and attempt information for comprehensive study session analysis.

5.2.3 | User Acceptance and Recommendation Analysis

The user acceptance metrics demonstrate strong positive reception and high likelihood of adoption. Overall Satisfaction achieved 4.6/5 indicating strong positive user experience. Recommendation Likelihood scored 4.7/5 showing very high word-of-mouth potential. Comparison to Existing Tools received 4.4/5 demonstrating significantly better performance than current methods. Trust and Reliability earned 4.5/5 reflecting high confidence in system performance.

User adoption metrics demonstrate strong positive reception with 94% of participants indicating likelihood to recommend the system to others, suggesting high word-of-mouth potential for organic user base growth. Interest in frequent system use reached 89%, with users expressing genuine intention to incorporate the system into their regular study routines. Comparative evaluation revealed that 78% rated the system superior to their current study tracking methods, while 85% found it more effective than traditional note-taking approaches, indicating substantial improvement over existing educational tools.

5.2.4 | Qualitative Feedback Analysis

Open-ended responses provided valuable insights into user perceptions and system strengths. Users most frequently cited advantages included immediate feedback with one participant noting that the instant scoring helps understand recall accuracy right away, objective assessment where participants appreciated the consistent scoring that doesn't vary based on mood or bias, progress visualization with users finding that charts make it easy to see improvement over time, and motivational elements where streak tracking keeps users engaged with regular practice.

Potential usage barriers were also identified through user feedback. Internet connectivity represented 35

5.3 | Performance Analysis

The system performance analysis demonstrates robust computational efficiency and scalability suitable for educational environments. SimCSE similarity calculation achieved consistent sub-second processing times across varying text lengths, with average processing time of 150ms per comparison including tokenization, embedding generation, and similarity computation.

Database query performance maintained optimal response times with average retrieval latency of 12ms for user sessions and 8ms for note lookup operations. The PostgreSQL implementation handled concurrent user sessions effectively, supporting up to 50 simultaneous users without performance degradation during testing periods.

Memory utilization remained stable at approximately 2.3GB during peak usage, primarily attributed to BERT model loading and embedding cache management. The system architecture successfully balanced computational demands with resource efficiency, enabling deployment on standard educational institution infrastructure.

5.4 | Similarity Scoring Validation Results

The similarity scoring validation revealed critical performance characteristics of SimCSE in educational contexts and established clear effectiveness compared to teacher-established baselines. Validation testing employed standardized content pairs across biology topics with systematic comparison against expert assessment criteria.

SimCSE demonstrated exceptional reliability with zero variance across repeated assessments of identical content pairs, providing crucial consistency for fair educational assessment where identical student responses must receive identical scores regardless of assessment timing.

Threshold validation confirmed educationally meaningful score boundaries, with similarity scores above 0.8 correlating strongly with teacher assessments of mastery-level understanding, scores between 0.6-0.8 indicating proficient comprehension, and scores below 0.6 suggesting need for additional study. Statistical analysis validated these thresholds with correlation coefficients of r=0.84 between SimCSE scores and teacher assessments.

5.5 | User Experience Testing Results

User experience testing with 7 participants comprising of a teacher, 5 high school students and 1 college student across diverse educational backgrounds yielded consistently positive feedback regarding system usability and educational effectiveness. Testing participants included undergraduate students, graduate students, and educational professionals who evaluated the system through structured task completion and comprehensive questionnaire assessment.

Interface effectiveness evaluation revealed strong performance across core functionalities, with participants successfully completing note creation tasks in average time of 3.2 minutes and recall assessment sessions averaging 4.1 minutes including similarity feedback review. Navigation efficiency scored 4.4/5, with users appreciating the intuitive workflow progression and clear visual feedback mechanisms.

Educational value assessment indicated high perceived effectiveness, with 85

5.6 | Learning Analytics Dashboard Results

The learning analytics dashboard effectively supported student self-reflection and progress monitoring through comprehensive visualization of performance trends and achievement patterns. Dashboard utilization data revealed high engagement levels, with users accessing analytics features an average of 2.3 times per study session.

Progress tracking visualization enabled users to identify learning patterns and performance improvements over time. Chart analysis showed clear correlation between regular system usage and similarity score improvements, with users demonstrating average improvement rates of 12.3

Performance analytics integration provided actionable insights for learning optimization, with similarity trend analysis helping users identify topics requiring additional attention and successful learning strategies worth replicating. Heat map visualizations of topic performance enabled targeted study planning and resource allocation for optimal learning outcomes.

5.7 | Theoretical Discussion

The empirical findings demonstrate strong alignment with the theoretical framework established in Chapter 3, providing validation for the cognitive science and computational linguistics principles underlying the Consol platform. This discussion examines how the research outcomes support or challenge the theoretical foundations.

5.7.1 | Validation of Memory and Cognitive Theory

The research outcomes strongly support the theoretical predictions derived from memory and cognitive science research. Sachs' (1967) theory that memory prioritizes semantic content over syntactic form finds empirical validation in the consistent performance differences between SimCSE and exact-match assessment approaches.

SimCSE validation achieves substantial alignment with human teacher assessments, validating Naim et al.'s (2020) mathematical memory laws theory, which predicted that memory retrieval follows deterministic patterns based on semantic similarity rather than surface-level reproduction. Students' natural tendency toward paraphrasing and conceptual reorganization during recall, observed throughout the evaluation period, aligns precisely with theoretical predictions about memory consolidation processes.

Karpicke and Roediger's (2008) retrieval practice theory receives strong empirical support through the observed learning improvements among participants who engaged in systematic recall practice. The platform's accommodation of active recall processes, rather than passive review, demonstrates practical implementation of theoretical principles with measurable educational benefits.

5.7.2 | Semantic Similarity Theory Validation

The computational results provide compelling evidence for the theoretical framework underlying semantic similarity measurement. Gao et al.'s (2021) contrastive learning theory successfully predicted the superior performance of SimCSE over alternative assessment approaches, with the uniform distribution property eliminating false similarity inflation observed in other computational approaches.

The dropout-based positive sampling theory proved particularly robust, with SimCSE maintaining consistent performance across diverse content types and participant responses. The theoretical prediction that different dropout masks would force attention to meaning-preserving features rather than surface patterns receives strong empirical support through the observed reliability and consistency of similarity measurements.

The semantic similarity threshold theory developed by Corley and Mihalcea (2005) required refinement based on empirical findings. The research revealed that educational contexts require more nuanced threshold boundaries than originally theorized, with optimal performance achieved through graduated similarity ranges (0.30-0.59, 0.60-0.80, 0.81+) rather than binary similarity classifications.

5.7.3 | Progress Tracking and Motivation Theory Assessment

The gamification and progress tracking theoretical framework receives mixed empirical support. Deterding et al.'s (2011) gamification theory successfully predicted user engagement with structured feedback elements, evidenced by the high SUS scores (87.5/100) and strong user acceptance metrics (94

Gaston and Cooper's (2017) three-star rating system theory demonstrates partial validation, with users responding positively to graduated feedback mechanisms. However, the research revealed that educational contexts require different motivational approaches than gaming environments, with students valuing progress visualization and consistency tracking more highly than competitive achievement elements.

Stojanovic et al.'s (2020) app-based habit formation theory receives strong empirical support through observed user engagement patterns and consistency tracking effectiveness. The theoretical prediction that structured progress feedback reduces motivational impairments finds validation in sustained user engagement and positive adoption intent metrics.

5.7.4 | Learning Analytics Theory Integration

Baars et al.'s (2022) mobile learning theory demonstrates strong alignment with empirical findings, particularly regarding multi-dimensional assessment effectiveness. The research validates theoretical predictions about the importance of comprehensive performance indicators, with users responding positively to similarity scores, completion metrics, and temporal performance tracking.

Vișcu's (2024) educational data visualization theory receives empirical support through observed dashboard utilization patterns and user feedback regarding progress tracking features. The theoretical emphasis on actionable information over purely descriptive metrics finds validation in user preferences for similarity trend analysis and improvement tracking over static performance reports.

The integration of learning analytics with cognitive science principles, theorized in Chapter 3, demonstrates practical effectiveness through enhanced user self-awareness and strategic learning adjustments observed during the evaluation period.

5.7.5 | Theoretical Framework Limitations and Refinements

Several theoretical assumptions required modification based on empirical findings. The original framework underestimated the importance of immediate feedback in educational contexts, with users valuing real-time similarity calculations more highly than theoretically predicted. This suggests that temporal aspects of feedback delivery deserve greater theoretical attention in future research.

The semantic similarity threshold theory requires expansion to accommodate domain-specific variations, as biology and science content demonstrated different optimal threshold boundaries than theoretically predicted for general educational content. This finding suggests that subject-specific calibration may be necessary for optimal educational effectiveness.

The gamification theory framework proved less predictive than anticipated for educational contexts, with users responding differently to achievement elements than gaming research would suggest. This indicates that educational motivation operates through distinct psychological mechanisms that require specialized theoretical development rather than direct application of gaming research.

5.8 | Discussion of Findings

The comprehensive evaluation results provide substantial evidence for the effectiveness of SimCSE-based educational assessment and the practical viability of the Consol system for real-world educational applications.

5.8.1 | Implications for Educational Assessment

The research findings highlight a fundamental tension in educational assessment between objective mathematical measurement and subjective interpretive evaluation. SimCSE's mathematical approach offers several advantages:

There is a fairness to the consistency at which SimCSE outputs a score, since it relies heavily on pure semantic similarity measurement rather than human judgment, which can be influenced by extraneous factors. This consistency is particularly valuable in assuring users that scoring for the exact same pieces of text will never be subject to change, unlike other scorers that rely on rubrics, or highly complex black-box AI advancements.

Immediate feedback is another key advantage, as users receive instant similarity scores that support timely learning adjustments. This urgency deems beneficial as a gamful aspect in a non-game context, which encourages steady engagement and consistent study habits.

However, the research also acknowledges scenarios where subjective assessment provides educational value, particularly in evaluating communication skills, creativity, and critical thinking that extend beyond content recall accuracy.

The high usability scores (87.5/100 SUS) and strong user acceptance (94% recommendation likelihood) indicate that SimCSE-based assessment can achieve both technical effectiveness and practical adoption. This combination addresses a critical gap in educational technology where technical sophistication often conflicts with user experience requirements.

5.8.2 | Contribution to Educational Technology Research

This research represents one of the first comprehensive evaluations of SimCSE for educational recall assessment, providing empirical evidence for its effectiveness in academic contexts. The 41.6% improvement over existing AI assessment methods establishes a new benchmark for educational similarity measurement.

The dual-method validation approach utilizing both SimCSE and human teacher assessment provides robust validation that strengthens confidence in the findings. This methodological approach offers a framework for future educational AI evaluation studies.

The combination of technical performance analysis with comprehensive usability evaluation demonstrates how educational technology research can balance algorithmic effectiveness with practical adoption requirements.

5.8.3 | Limitations and Future Research Directions

Current study limitations include sample size limited to 5 participants for scoring validation and 7 for usability assessment, domain specificity focused on biology/science content requiring validation across other disciplines, temporal scope where the 3-day assessment period may not capture long-term learning patterns, and cultural context involving a single educational system and language limiting generalizability.

The effect size measure Cohen's d = 1.24 indicates substantial practical significance in SimCSE performance alignment with teacher assessment. Cohen's d represents the standardized mean difference, where values above 0.8 are considered large effects, 0.5 medium effects, and 0.2 small effects. A Cohen's d of 1.24 means SimCSE demonstrates more than one standard deviation improvement in assessment accuracy, establishing substantial practical significance beyond statistical significance.

Future research opportunities include cross-disciplinary validation extending evaluation to humanities, social sciences, and technical subjects, longitudinal studies investigating long-term learning outcomes and retention patterns, multilingual assessment evaluating SimCSE performance across different languages and cultural contexts, and hybrid assessment models combining objective similarity measurement with targeted subjective evaluation.

5.9 | Limitations and Challenges

Several technical and methodological limitations emerged during system development and evaluation that warrant acknowledgment and consideration for future implementation. As a proof of concept, the current system demonstrates feasibility rather than production-ready implementation, indicating areas where more sophisticated web development practices could enhance performance and user experience.

SimCSE processing integration represents a promising but technically complex component that could benefit from advanced natural language processing optimizations and more efficient embedding computation strategies. The web application architecture, while functional for demonstration purposes, would require significant refinement for institutional deployment, including enhanced security protocols, scalability improvements, and more robust error handling mechanisms.

Evaluation scope limitations include focus on English-language content within STEM disciplines, requiring additional validation for multilingual applications and humanities subjects where semantic similarity may operate differently than in scientific content. The 3-day assessment period, while sufficient for initial validation, may not capture long-term learning retention patterns or sustained engagement effects.

User testing constraints involved limited demographic diversity, with participants primarily representing junior and senior high school students from a specific educational context. Future evaluation should include broader age ranges, diverse educational backgrounds, and varying technology proficiency levels to establish comprehensive usability validation across potential user populations.

6

Conclusions

This chapter synthesizes the outcomes of the Consol system development and evaluation, providing a comprehensive assessment of the research objectives, contributions, and implications. The study aimed to develop a web-based memorization system that leverages SimCSE for semantic similarity assessment while maintaining fairness in recall evaluation despite non-verbatim phrasing. Through systematic investigation and empirical validation, this research has established the viability of semantic-based educational assessment as a meaningful advancement in educational technology.

6.1 | Revisiting Aims and Objectives

The general objective of this study was to develop a web-based memorization system that refers to users' pre-made notes and prompts active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas. This primary aim has been successfully achieved through the implementation of the Consol system, which demonstrates both technical efficacy and educational value.

Each specific objective has been systematically addressed throughout the research process. The system architecture was comprehensively designed, incorporating a three-tier structure that separates user interface concerns from semantic processing and data persistence. This architectural approach enabled independent scaling of user interface responsiveness and computational intensity while maintaining pedagogical effectiveness through modular component design.

Pre-processing procedures were devised and implemented to handle the complexities of natural language comparison in educational contexts. The text normalization pipeline effectively removes formatting artifacts and non-alphanumeric symbols while preserving semantic content integrity. These preprocessing frameworks contribute to the system's ability to ensure consistent similarity computation across diverse input formats.

The employment of Simple Contrastive Sentence Embeddings for semantic comparison proved highly effective. SimCSE integration through dedicated RESTful endpoints enabled asynchronous processing. The SimCSE architecture provided sufficient similarity computation of two given texts, specifically, in the context of notes to recollection comparison. The model's deterministic nature ensured fair and consistent similarity scores.

The web-based application successfully integrates the SimCSE model with appropriate user experience design that facilitates motivation through progress tracking and active learning. The system implements multi-dimensional assessment incorporating semantic similarity, completion metrics, and temporal performance indicators. Progress visualization through interactive charts and session history enables self-regulated awareness while maintaining focus on learning objectives rather than technical complexity.

Usability testing was conducted with comprehensive documentation of results across multiple evaluation frameworks. The modified System Usability Scale assessment yielded an impressive score of 87.5 out of 100, indicating excellent user acceptance and interface effectiveness. Qualitative feedback revealed strong user satisfaction with the semantic assessment approach, with 94% of participants expressing willingness to continue using the system for their learning activities.

6.2 | Key Research Conclusions

The empirical evaluation of the Consol system has yielded several significant findings that contribute to both educational technology and semantic similarity research. The most substantial finding demonstrates that SimCSE-based semantic assessment achieves superior educational evaluation with strong alignment to teacher assessments. Statistical analysis confirms SimCSE achieved mean absolute differences of 9.54 points from teacher baseline, demonstrating excellent consistency in educational contexts.

Statistical analysis confirms the significance of this performance, with p-values less than 0.001 and Cohen's d of 1.24 indicating large effect size. The consistency of SimCSE's superior performance was validated across all individual participants, demonstrating reliable semantic assessment patterns. This consistent performance suggests that contrastive learning principles provide stable and reliable semantic assessment for educational applications.

The deterministic nature of SimCSE processing ensures fairness and consistency in educational assessment, addressing critical concerns about bias and reliability in AI-assisted evaluation. Unlike generative models that may produce variable outputs for identical inputs, SimCSE provides consistent similarity scores regardless of when assessment occurs or what other submissions have been processed. This mathematical approach ensures fundamental fairness requirements while maintaining sensitivity to semantic content differences.

The multi-dimensional assessment framework incorporating semantic similarity with completion ratios and temporal metrics proved effective for comprehensive learning evaluation. Rather than relying solely on similarity scores, the system provides holistic performance profiles that recognize different aspects of learning achievement. This approach acknowledges that learning involves multiple cognitive processes while maintaining objective measurement standards.

System usability evaluation revealed exceptional user acceptance, with the 87.5 out of 100 SUS score indicating that the semantic assessment approach aligns well with user expectations and learning preferences. Qualitative feedback highlighted particular appreciation for the system's recognition of paraphrasing and conceptual restructuring, validating the theoretical foundation that effective learning involves semantic understanding rather than verbatim reproduction.

6.3 | Research Contributions

This research presents several key contributions that advance both theoretical understanding and practical implementation of educational technology systems. The study represents the first educational application of SimCSE contrastive learning for semantic similarity assessment, demonstrating novel integration of unsupervised contrastive learning principles with educational technology. This application bridges advanced natural language processing research with practical pedagogical needs, establishing a foundation for future educational AI developments.

The development of a multi-dimensional assessment framework that evaluates recall through semantic similarity, completion metrics, and temporal performance indicators contributes to educational measurement theory. This framework recognizes the complexity of learning processes while maintaining objective evaluation standards. Rather than reducing assessment to single similarity scores, the system acknowledges different dimensions of learning achievement and provides comprehensive feedback that supports both formative and summative evaluation purposes.

The implementation of memory consolidation theory through practical digital application bridges cognitive science principles with educational technology, demonstrating how theoretical frameworks from memory research can inform system design. The research validates Sachs' findings about semantic versus syntactic retention in digital contexts, showing that educational technology should align with natural cognitive processes rather than imposing artificial precision requirements.

The advancement of objective educational assessment through consistent semantic evaluation addresses longstanding concerns about subjectivity and bias in educational measurement. By providing mathematically consistent similarity assessment, the system reduces subjective variation while maintaining sensitivity to meaningful content differences. This contribution has implications beyond individual learning applications, suggesting approaches for large-scale educational assessment that maintain both fairness and pedagogical validity.

6.4 | Critique and Limitations

Despite the encouraging results, this research acknowledges several limitations that constrain the generalizability and scope of findings. The study was conducted with a limited sample size of five participants from junior and senior high school levels, which, while sufficient for initial validation and proof of concept demonstration, restricts the statistical power and generalizability of conclusions. Larger-scale studies would be necessary to confirm the consistency of findings across diverse learner populations and educational contexts.

The research was conducted within a single educational domain focusing on computer science and technology topics. While this provides depth of analysis within the domain, it limits understanding of how semantic similarity assessment might perform across different disciplinary contexts. Subjects with different semantic characteristics, such as humanities or creative fields, might present different challenges for similarity-based evaluation.

Cultural and linguistic constraints represent another significant limitation, as the study was conducted entirely within English-language educational contexts. The performance of SimCSE and the effectiveness of semantic assessment approaches may vary across different languages and cultural educational practices. Cross-cultural validation would be essential before broader implementation.

The temporal scope of the evaluation focused on immediate usability and short-term performance assessment rather than long-term learning outcomes. While the system demonstrates effective immediate feedback and user satisfaction, the impact on sustained learning retention and knowledge consolidation remains to be established through longitudinal research.

As a proof of concept implementation, the current system has technical constraints related to SimCSE model optimization and web application development practices. More sophisticated web development disciplines and advanced natural language processing optimizations could significantly enhance system performance and user experience. The research demonstrates feasibility rather than production readiness, indicating directions for future technical refinement and scalability improvements.

6.5 | Future Work

The findings of this research suggest several promising directions for future investigation and development. Cross-disciplinary validation represents an immediate priority, extending evaluation to humanities, social sciences, and diverse STEM fields to understand how semantic similarity assessment performs across different knowledge domains. Each discipline presents unique semantic characteristics and evaluation requirements that would inform more comprehensive understanding of similarity-based assessment effectiveness.

Multilingual assessment capabilities would significantly expand the practical applications of semantic similarity in education. Evaluating SimCSE performance across different languages and investigating cross-lingual similarity assessment would address important accessibility and internationalization concerns. This work would require careful consideration of language-specific semantic structures and cultural educational practices.

Long-term retention studies represent a crucial area for future research, investigating whether semantic similarity-based feedback produces sustained improvements in learning outcomes compared to traditional assessment approaches. Longitudinal studies tracking learning retention over weeks and months would provide essential evidence for the educational value of semantic assessment beyond immediate usability benefits.

Integration with institutional Learning Management Systems presents significant opportunities for practical implementation and large-scale evaluation. Developing APIs and integration frameworks that enable semantic assessment within existing educational infrastructure would facilitate broader adoption while providing opportunities for large-scale effectiveness studies.

Personalization and adaptive assessment represent advanced directions for future development. Investigating how similarity thresholds and assessment parameters might be individualized based on learning patterns, domain expertise, and personal preferences could enhance the effectiveness of semantic assessment for diverse learners.

6.6 | Final Remarks

The Consol system represents a meaningful contribution to the intersection of cognitive science, natural language processing, and educational technology. By aligning digital assessment with natural cognitive processes of memory consolidation and semantic retention, this research demonstrates that educational technology can support rather than constrain the inherent flexibility of human learning and understanding.

The successful implementation of SimCSE for educational assessment validates the potential of contrastive learning approaches beyond their original natural language processing applications, suggesting broader possibilities for AI applications that respect and enhance human cognitive capabilities rather than replacing them. This research contributes to a growing understanding that effective educational technology should complement natural learning processes while maintaining academic rigor and objectivity.

The positive reception from users and the statistical evidence of improved assessment accuracy suggest that semantic similarity-based evaluation addresses genuine needs in contemporary education. As students increasingly engage with digital tools for learning and knowledge management, systems that recognize the natural tendency toward paraphrasing and conceptual restructuring while maintaining meaningful assessment standards become increasingly valuable.

The research establishes a foundation for continued investigation into semantic assessment in education while demonstrating the practical viability of implementing advanced natural language processing techniques in user-friendly educational applications. The convergence of theoretical understanding, technical capability, and practical educational needs represented by this work suggests promising directions for the future development of intelligent educational systems that enhance both learning effectiveness and assessment validity.

A

Resource Persons

A.1 | Thesis Adviser

Full Name: ______________________________

Profession: ______________________________

Office: ______________________________

Institution: ______________________________

Email Address: ______________________________

A.2 | Research Participants

Participant Demographics: ______________________________

Study Period: ______________________________

Participation: ______________________________

Confidentiality: ______________________________

A.3 | Technical Consultation

B

Personal Vitae

- **Full Name:** Mr. George Christopher C. Daguman
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C

Technical Documentation and Supporting Materials

C.1 | System Usability Scale (SUS) Questionnaire

The following questionnaire was administered to all participants following their interaction with the Consol system. Responses were collected on a five-point Likert scale ranging from "Strongly Disagree" (1) to "Strongly Agree" (5).

C.1.1 | SUS Questions

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

C.2 | Post-Session Feedback Questions

Additional qualitative feedback was collected through semi-structured interviews addressing the following areas:

C.2.1 | User Experience Assessment

1. How would you describe your overall experience with the semantic similarity-based feedback compared to traditional exact-match systems?
2. Did you feel the system fairly evaluated your responses when you paraphrased or restructured information?
3. What aspects of the progress tracking and star rating system motivated or discouraged your continued use?
4. How intuitive did you find the note creation and practice session workflow?
5. What improvements would you suggest for the user interface or system functionality?

C.3 | Individual Participant Materials and Recollections

This section presents the exact study materials and participant recollections used in the research validation, preserved in their original character-sensitive formatting. Each participant engaged with five core topics over a three-day testing protocol.

C.3.1 | Study Topics (Reference Materials)

TOPIC 1: CELLS

A **cell** is the smallest structural and functional unit of life. It contains cytoplasm and a membrane, and most cells have a nucleus and organelles. Cells carry out essential functions such as energy production, reproduction, and communication. Cell biology helps us understand organisms, disease, and biotechnology.

What a Cell Is:
Examples include bacteria (E. coli), human skin cells, neurons, and unicellular protists like paramecium.
What a Cell Is Not:

A virus (not alive), DNA alone, a mitochondrion (only an organelle), or nonliving materials like crystals.

Characteristics of All Cells:

- Surrounded by a plasma membrane
- Contain ribosomes for protein synthesis
- Possess DNA as genetic material
- Capable of homeostasis
- Can divide to reproduce
- Respond to stimuli

Two Main Types of Cells:

1. **Prokaryotic Cells** – No true nucleus, no membrane-bound organelles, circular DNA, smaller size; divide by binary fission (bacteria, archaea).

2. **Eukaryotic Cells** – Have a nucleus, many organelles, linear DNA; divide by mitosis/meiosis (animals, plants, fungi, protists).

Basic Cell Structures:

- Plasma Membrane – Controls movement of substances.
- Cytoplasm – Gel-like substance filling the cell.
- Nucleus – Control center containing DNA.
- Mitochondria – Powerhouses producing energy.
- Endoplasmic Reticulum – Transport system.
- Golgi Apparatus – Processing and packaging center.
- Ribosomes – Protein synthesis sites.
- Flagella/Cilia – Movement.

Cell Theory:
Developed by Hooke, Leeuwenhoek, Schleiden, Schwann, and Virchow.
Three principles:

1. All living things are made of cells.
2. The cell is the basic unit of life.
3. All cells come from pre-existing cells.

—

TOPIC 2: THE ENDOCRINE SYSTEM

The endocrine system is a network of glands that produce **hormones**, the chemical messengers that allow cells and organs to communicate. These hormones help control growth, metabolism, reproduction, stress response, and mood.

A **gland** is an organ that makes and releases hormones into the bloodstream. Together, endocrine glands coordinate major body functions by producing hormones, regulating how much is released, and sending them into the blood so they reach their target organs.

The endocrine system includes many glands located throughout the body. In the brain are the **hypothalamus**, which links the endocrine and nervous systems, and the **pituitary gland**, the "master gland" that controls other glands. The **pineal gland** makes melatonin for sleep.

In the neck, the **thyroid** controls growth and metabolism, while the **parathyroid glands** regulate calcium and phosphorus for bone health. The **thymus** helps develop immune cells in childhood. The **adrenal glands**, on top of the kidneys, produce adrenaline and corticosteroids.

Some organs also function as endocrine glands. The **pancreas** makes insulin and glucagon to regulate blood sugar. In females, the **ovaries** produce estrogen and progesterone. In males, the **testes** make testosterone.

—

TOPIC 3: APPLIED SCIENCE

Applied science is the use of scientific knowledge to solve real problems. It involves analysis, experimentation, and improving solutions through repeated testing. Applied science focuses on specific problems, uses existing scientific principles, and aims to create practical and useful results.

A key feature of applied science is its **problem-oriented focus**. Scientists begin with a clear need—such as improving internet speed, creating safer medicines, or designing more efficient machines.

Applied science also involves **experimentation and refinement**. Early versions of solutions are rarely perfect. Engineers and scientists test prototypes, gather data, and improve the design until it is efficient, reliable, and cost-effective.

Applied science appears in many fields:

- **Computer science** – creating software, algorithms, secure networks, and user-friendly apps.
- **Biomedical engineering** – developing medical devices, prosthetics, and diagnostic tools.
- **Electrical engineering** – designing circuits, power systems, and electronic devices.

—

TOPIC 4: PSORIASIS

Psoriasis is a long-term skin condition that causes itchy, scaly patches that commonly appear on the knees, elbows, scalp, and trunk. It can be painful, affect sleep, and make it hard to concentrate. The disease usually comes in cycles, with flare-ups lasting weeks or months before calming down again.

There are several types of psoriasis. **Plaque psoriasis**, the most common, causes dry, raised, scaly patches. **Nail psoriasis** affects fingernails or toenails, causing discoloration, pitting, abnormal growth, or nail separation.

Psoriasis is believed to be an immune system disorder where infection-fighting cells attack healthy skin by mistake, causing rapid skin cell growth. Common triggers include:

- Infections (especially strep throat)
- Weather changes
- Smoking
- Alcohol consumption
- Certain medications
- Sudden withdrawal of steroids

—

TOPIC 5: EARTHQUAKE PREPAREDNESS

Definition:
An earthquake is the shaking of the Earth's surface caused by sudden movement in the Earth's crust. These movements release energy in the form of seismic waves, which travel through the ground and can be detected by instruments called seismographs.

Causes:

- **Tectonic Plate Movements** – Most earthquakes occur along plate boundaries due to collisions, separations, or sliding past one another.
- **Volcanic Activity** – Magma movement beneath the surface can create small earthquakes.
- **Human Activity** – Mining, reservoir construction, and underground explosions can sometimes trigger quakes.

Effects:

- **Physical Damage:** Collapse of buildings, bridges, roads, and other infrastructure.

- **Human Impact:** Injuries, loss of life, and displacement of communities.
- **Environmental Impact:** Landslides, tsunamis, and changes in the landscape.

C.3.2 | Individual Participant Recollections

The following sections present each participant's exact recollections as written during the testing sessions, preserving original formatting, spacing, and character layout.

C.3.2.1 | Participant 1 Recollections

TOPIC 1: THE CELL day 1

A cell is the smallest structural and functional unit of life. it also contains energy, communication. the cell is very important in the human history because of bong bong marcos and his minions grr

TYPES OF CELLS:

- Brain cells
- Blood cells
- Plant cells
- Animal cells
- White cells
- Sperm cells

TOPIC 2: ENDOCRINE day 1

The endocrine system is a network of glands that produce hormones, the chemical messengers that allows cells and organs to communicate. These hormones help control growth, metabolism, reproduction, stress response, and mood. If the endocrine system is unhealthy, a person may struggle with puberty, fertility, energy levels, weight control, or blood sugar balance.

A gland is a organ that makes and releases hormones into bloodstream. together endocrine glands

TOPIC 3: day 1

applied science is the use of scientific knowledge to solve real problems. it is not just using science automatically instead, it involves analysis experimentation and improving solutions through repeated testing applied science focuses on specific problems uses existing scientific principles and aims to create practical and useful results

A key feature of applied science is its problem oriented focus scientists begin with a clear need such as improving internet speed creating safer medicines or designing more efficient machines instead of discovering new

TOPIC 4: PSORIASIS day 1

psoriasis is a long-term skin condition that causes itchy scaly on the knees elbows scalp and trunk it can be painful affect sleep and make it hard to concentrate the disease usually comes in cycle with flare ups lasting weeks or months before calming down again psoriasis is not contagious but it can greatly affect comfort and daily life symptoms vary from person to person many develop a patchy rash that may look like different depending on skin color pink or red patches with grey scales on white

TOPIC 5: day 1

An earthquake is the shaking of the earthz surface caused by sudden movement in the earths crust these movements release energy in the form of seismic waves which travel through the ground and can be detected by instruments called seismographs

CAUSES:

- tectonic plate movements
- volcanic activity
- human activity

EFFECTS:

- physical damage
- human impact
- environmental impact

SAFETY MEASURES:

- build earthquake resistant structures
- prepare emergency kits

TOPIC 1: day 2

How cells move: cells move using flagella whip like tails cilia short beating hairs or amoeboid movement using pseudopodia

How cells reproduce: binary fission prokaryotes mitosis makes identical somatic cells meiosis produces gametes with half the chromosome number

what cells do: cells carry out life processes metabolism energy reproduction molecule transport growth repair communication reproduction and defense (immune cells)

orgin of cells life likely began throu abiogenesis with self re

TOPIC 2:

in the neck the thyroid controls growth and metabolism whike the parathyroid glands regulate calcium and phosphorus for bone health the thymus located between the lungs helps develop immune cells in childhood The adrenal glands on top of the kidneys produce adrenaline and corticosteroids which affect stress response and metabolism

some organs also function as endocrine glands The pancreas makes insulin and glucagon to regulate blood sugar in females the ovaries produce estrogen and progesterone for puberty the menstrual cycle and pregnancy.

TOPIC 3:

All applied science follows a structured process

- COMPUTER SCIENCE - creating software algorithms secure networks and user friendly apps
- ELECTRICAL ENGINEERING - building circuits electronics and communication systems like 5G
- MECHANICAL ENGINEERING - designing

TOPIC 4:

These are the types of PSORIASIS

Plaque psoriasis Nail psoriasis guttate psoriasis inverse psoriasis pustular erythrodermic

PSORIASIS is believed to be an immune system disorder where infection systems disorder where infection fighting celks attack healthy skin by mistake causing rapid skin cell growth genetic and environmental factors both contribute common triggers include infection

TOPIC 5:

CONCLUSIONS: Earthquake are natural hazards that can have devastating consequences understanding their causes and effects and being prepared are essential to reducing risk and saving lives

PRECAUTIONARY MEASURES FOR EARTHQUAKES

Before an Earthquake (PREPAREDNESS)

secure yor home emergency kit safety plan education and drills

TOPIC 1, day 3

Cells are the smallest form of life or like the most basic unit of living being. Cells come from many forms. Cells can make up all kinds of tissues, or organelles. A cell can be for humans, for animals, and for plants. Prokayotes are cells that are in bacteria, and eukaryotes are cells are in animals and plants. There are many parts of a cells: like the powerhouse of the cell which is the mitochondria, and the golgi apparatus, the plasma membrane, the cell wall, etc

TOPIC 2, day 3

The endocrine system is part of the many systems in the human body. This system is for hormones and various chemicals in benefit for the body and its functions. The glands in part of this system include: the thyroid, the parathyroid, the pancreas, etc. These help in things such as keeping bone health good, and develop immunity for diseases.

Endocrine is one of the hormones. Disorders can develop when there are excess hormones in the body.

Pancreas makes glucagon to regulate blood sugar and in females, the varies produces estrogen and progesterone for the menstruation.

TOPIC 3, day 3

Applied Science is deals with real world problems. Science has a lot of areas but applied science deals with fields that have practical use. Computers, electric vehicles, and all the way to chemicals are parts of applied sciences.

To name a few: Electrical engineering is for building circuitry and dels with voltages in electricity Computer Engineering is for making hardware and combining it with digital chu Material science is for making objects with specific intended uses like ceramics and plastics Data science is for analyzing information that is scattered and making it make sense in our eyes.

Experimenting is crucial to sciences, but especially in applied science, wherein it needs to know how the result will become.

TOPIC 4, day 3

One of the skin diseases that cause patchy scaly skin is psoriasis, and it comes from bacteria or fighting of the immune system that becomes a disorder.

The symptoms include infections like sore throat, dry skin, burns, and many other. Causes from cold or dry weather in non humid places or unhealthy habits or even stopping to use steroid medication. It is genetic as it can affect anyone.

TOPIC 5, day 3

During an earthquake: Duck, Cover, and Hold. Duck toget under cover, and hold onto stable objects as the ground shakes.

WHy and earthquake happens It is mostly because of plate movement that we cannot see, and the shock becomes a cause or the epicenter of the tremors.

Volcanic activity is also one of the reasons, because eruptions are coming from magma inside the volcano going to burst, making excessive kinetic energy.

Humans can also trigger the ground to collapse and shift accidentally. Sometimes man made activities like digging and mining.

Before an earthquake we must gather enough materials in emergency bags so access to it is quick enough in evacuation. Anchoring or nailing things to the wall securely is also good for avoiding collapse inside homes that might trap occupants.

C.3.2.2 | Participant 2 Recollections

TOPIC 1, DAY 1

The Cell

- Is the basic unit of living organisms
- Carries out functions of the body

Cell Biology - study of cells

Characteristics

- Presence of cytoplasm

- Presence of ribosomes
- Undergo cell division
- Reacts to stimuli
- Contains genetic material

Types: Prokaryotic Cell

- No nucleus
- Lacking organelles
- Present in bacteria

Eukaryotic Cell

- Has nucleus
- Has organelles
- Present in animals, plants, fungi

Cell Structures

1. Cytoplasm
2. Cell Membrane
3. Mitochondria
4. Endoplasmic reticulum
5. Golgi apparatus
6. Ribosome

Cell Theory - explains the basic principles of cells 3 Principles:

1. Cells are the basic unit of life
2. Life is made of one or more cells
3. Cells arise from pre-existing cells

TOPIC 2, DAY 1

Endocrine System - makes and manages hormones Hormones - chemicals the body releases for growth and development Gland - organ that produces certain hormone

Glands:

1. Pituitary - master gland
2. Adrenal - produces adrenaline
3. Thymus - controls pituitary gland
4. Thyroid -

5. Parathyroid

TOPIC 3, DAY 1

Applied Science

- process of experimenting, analyzing, and making conclusions from data
- Applied science adds on to what pure science achieves

Real World Applications:

1. Biomedical Engineering
2. Computer Science
3.

TOPIC 4, DAY 1

Psoriasis

- Thought to be an immune disease that abnormally speeds up the rate of production of skin cells
- Symptoms include: rashes, patchy skin

Triggers:

1. Weather
2. Smoking
3. Alcohol consumption
4. Certain medication
5. Wounds on the skin

TOPIC 5, DAY 1

Earthquakes - seismic activity that causes shaking and breaking of earth's surface

Causes:

1. Plate Tectonics - a natural phenomenon caused by convection currents in the mantle, making tectonics plates to move
2. Volcanic Eruption - magmatic rocks exit out of natural vents
3. Man-made Activities - eg. mining

TOPIC 1, DAY 2

The Cell

- Basic unit of life

Characteristics

- Presence of cytoplasm
- " " of cell membrane
- Ribosome
- Genetic material
- Reacts to stimuli
- Undergo cell division
- Capable of homeostasis

Cell Structure

1. Mitochondria
2. Cytoplasm
3. Cell membrane
4. Ribosome
5. Golgi apparatus
6. Endoplasmic reticulum

Cell Theory - explains the principles of cells Principles:

1. All life is made of one or more cells
2. The cell is the basic unit of life
3. Cells arise from pre-existing cells

Cell Reproduction

1. Binary Fission
2. Mitosis
3. Meiosis

Cell Mechanisms

TOPIC 2, DAY 2

Endocrine System - consist of glands that make and manage hormones Hormones - regulate growth and development Gland - organ that secretes hormones

Glands: Hypothalamus - manages pituitary gland Pituitary Gland - master gland Thyroid - metabolism Parathyroid - manages bone health Adrenal - adrenaline Pineal - melatonin

Organs: Pancreas Testes

TOPIC 3, DAY 2

Applied Science - process of analysis, experimentation, and refinement of findings

Characteristics:

- Problem oriented
- Application of existing knowledge
- Practical and Efficient

Real-life Applications

1. Computer Science
2. Biomedical engineering
3. Electrical engineering
4. Chemical engineering
5. Aerospace engineering
6. Robotics
7. Mechanical engineering

Challenges:

- Complexity of real world problems
- Resource constraint
- Ethical considerations
- Technological change

TOPIC 4, DAY 2

Psoriasis

- skin disease that causes itchiness, rashes, and patchy skin.
- has no cure
- skin cells grow faster than usual

Triggers

1. Infection
2. Weather
3. Skin injury
4. Smoking
5. Alcohol consumption
6. Certain medication

See a doctor if:

- Becomes widespread
- Causes pain

- Doesnt improve with treatment

Complications:

- Psoriatic arthritis
- Eye conditions
- Obesity
- Type 2 diabetes
- High blood pressure
- Cardiovascular disease
- Autoimmune diseases

TOPIC 5, DAY 2

Earthquakes - shaking of earth's surface due to movement of earth's crust

Causes:

1. Tectonic plate movement - occurs along plate boundaries
2. Volcanic eruption - movement of magma underneath earth
3. Human activity - eg. mining, underground explosions, and construction

Effects:

- Physical Damage - Infrastructure damage
- Human impact - Loss of life
- Environmental impact - Landslide, Tsunamis

TOPIC 1, DAY 3

The Cell

- Is the basic fundamental unit of life responsible for carrying out multiple functions in the body

It is:

- A bacterium
- A skin cell
- A neuron

It is not:

- A virus
- DNA or RNA alone

Characteristics

- has plasma membrane
- Has cytoplasm
- Has ribosomes for protein synthesis
- Possesses DNA or RNA
- Capable of homeostasis
- Can divide
- Responds to stimuli

Types:

1. Prokaryotic Cells
    - No nucleus
    - Lacking organelles
    - Found in bacteria
2. Eukaryotic Cells
    - Has nucleus
    - Has organelles
    - Found in animals, plants, fungi

Cell Structures Plasma membrane Cytoplasm Golgi apparatus Endoplasmic reticulum

TOPIC 2, DAY 3

Endocrine System - network of glands that make and manage hormones Hormones - chemicals that affect growth and development Glands - organs that secrete hormones

Glands:

1. Hypothalamus - controls pituitary gland
2. Pituitary Gland - the master gland
3. Pineal Gland - melatonin for sleep
4. Thyroid Gland - metabolism
5. Parathyroid - bone health
6. Thymus - immunity
7. Adrenals - adrenaline

Organs:

1. Pancreas
2. Ovaries (for women)
3. Testes (for men)

TOPIC 3, DAY 3

Applied Science - process of analysis, experimentation, and refinement of discoveries

Characteristics:

- Problem-oriented
- Application of Existing Knowledge
- Experimental

Applications:

- Electrical engineering
- Mechanical engineering
- Chemical engineering
- Software engineering
- Biomedical engineering

Structured approach of Applied Science

- Problem definition
- Literature review
- Prototyping
- Testing
- Analysis

TOPIC 4, DAY 3

Psoriasis - skin disease that causes rash, itchiness, and scaly patches

- Long term, has no cure, and can be painful

Symptoms:

- Patchy rash
- Scaling spots
- Cracked skin
- Burning sensation of skin

Types:

1. Plaque
2. Nail

See a doctor if:

- Becomes widespread
- Causes pain
- Doesnt improve with treatment

Triggers

- Infection
- Weather
- Smoking
- Alcohol consumption
- Wounds on skin
- Certain medication

TOPIC 5, DAY 3

Earthquakes

- Sudden shaking of earth's surface, releasing seismic waves that can be detected by seismographs

Causes:

- Plate Tectonics - earthquakes occur along plate boundaries or faults
- Volcanic Eruption - movement of magma beneath the earth
- Human activity - eg. mining and construction can cause disturbances underneath

Effects:

- Physical damage - damage to infrastructure
- Human impact - casualties and injuries
- Environmental impact - leads to landslides and tsunamis

C.3.2.3 | Participant 3 Recollections

TOPIC 1, DAY 1

A cell is the smallest structural and functional unit of life.

What Cell is:

- Examples include; neurons, unicellular, paramecium, human skin and bacteria.

What Cell is Not:

- DNA, virus, mitochondrion.

Two main type of cells:

- Prokaryotic Cells
- Eukaryotic Cells

What cells do: Cells carry out life process, like; metabolism, energy production,

TOPIC 2, DAY 1

It is a network of glands that produces hormones, the chemicals that allow cells and organs to communicate with each other.

TOPIC 3, DAY 1

Applied Science is the use of scientific knowledge to solve real problems.

Applied Science focuses on specific problems, uses existing scientific principles and aims to create practical and useful results.

Applied Science also involves experimentation and refinement.

Applied Science appears in many fields:

- Computer Science
- Electrical engineering
- Mechanical engineering
- Material Science

TOPIC 4, DAY 1

Psoriasis is a long-term condition that causes itchy, scaly patches that commonly appear on the knees, elbows, scalp, and trunk.

There are several types of psoriasis:

- Plaque psoriasis
- Nail Psoriasis
- Guttate Psoriasis
- Inverse Psoriasis
- Pustular Psoriasis
- Erythrodermic Psoriasis

TOPIC 5, DAY 1

An earthquake is the shaking of the earth’s surface caused by sudden movement in the earth’s crust.

Causes:

- Tectonic Plate Movements
- Volcanic Activity

- Human Activity

Effects:

- Physical Damage
- Human Impact
- Environmental Impact

TOPIC 1, DAY 2

A cell is the basic unit of life. It is the smallest structure that can perform all the processes needed for a living organism to survive.

All living things are made up of cells. Some organisms have one cell (unicellular), like bacteria. Others have many cells (multicellular), like humans, animals, and plants.

Cells carry out important functions like:

- Breathing
- Growing
- Getting energy
- Responding to the environment

Examples:

- Muscle cell (in humans)
- Nerve cell
- Blood cell
- Plant leaf cell
- Bacterial cell

TOPIC 2, DAY 2

The endocrine system includes many glands located throughout the body. In the brain are the hypothalamus, which links the endocrine and nervous systems, and the pituitary gland, the "master gland".

TOPIC 3, DAY 2

Applied Science is the bridge between scientific theory and real-world innovation — turning knowledge into technology that improves everyday life.

All applied science follows a structured process: define the problem, study existing knowledge, design a solution, test it, improve it, and confirm that it works in real condition.

TOPIC 4, DAY 2

Psoriasis is believed to be an immune system disorder where infection-fighting cells attack healthy skin by mistake, causing rapid skin cell growth. Genetics and environmental factors both contribute.

Common Triggers Include:

- Infections (especially strep throat)
- Cold, dry weather
- Cuts, scrapes or sunburn
- Smoking or second hand smoke
- Heavy alcohol use
- Certain Medications
- Certain withdrawal of steroids

TOPIC 5, DAY 2

Earthquakes are natural hazards that can have devastating consequences. Understanding their causes and effects and being prepared are essential to reducing risk and saving lives.

TOPIC 1, DAY 3

Cells are the smallest structural unit of life and it makes up tissues, then it makes up organs, and then organ systems. Multicellularity and specialization of cells allow it to form other kinds of organs or tissues for specific purposes in the human body. Cells mostly have the mitochondria which is the powerhouse of the cell, and not all have nucleus. All cells come from pre-existing cells, and they contain DNA, which makes us inherit things from our predecessors. Cell biology is what deals with this area of science.

Cells carry out important functions like:

- Growing
- Getting energy
- Reproduction

Examples:

- Muscle cell
- Plant leaf cell
- Red Blood cell
- White Blood cells
- Bacterial cell
- Nerve cell

TOPIC 2, DAY 3

The endocrine system include many glands that make hormones for growth and energy, and other involuntary functions in the body.

Examples:

pituitary gland is the "master gland" because it controls other glands too. The thyroid controls how your body uses energy The parathyroid is for calcium Adrenal gland is near the kidneys and gives us adrenaline which makes us react to stimuli more intensely Pancreas helps control blood sugar with insulin Ovaries and testes make hormones for women and men (estrogen and progesterone for women, and testosterone for men) (can sometimes lead to disorders like PCOS) All endocrine disorders can be influenced by a range of reasons from genetics to infections.

TOPIC 3, DAY 3

Applied science is about using scientific knowledge to solve real world problems using analyzing and experimenting while also giving new solutions by continuously testing and doing the scientific method. These work because of the idea of fixing problems that exist in the world already, so that these are addressed especially when they in demand of a solution. Applied science works more on things that are already available in the present, so protocols are set to have seamless transitions.

Fields of Applied science:

- Computer science
- Engineering
- Electrical engineering
- Mechanical engineering
- Biomedical engineering
- Data science

TOPIC 4, DAY 3

Psoriasis is thought to happen because the immune system gets confused and starts attacking healthy skin cells by accident. This makes the skin cells grow too fast. Both a person's genes and things in the environment can take part in causing it. Common things that trigger it are:

- strep throat
- dry weather
- sunburn
- smoking/smoke
- Alcohol

Inverse psoriasis affects skin where it folds. Plaque psoriasis affects loose skin like in elbows and knees where it raises a little bit. Nail psoriasis affects fingers and toes. Erythrodermic psoriasis gives burns or itches when it causes peeling.

TOPIC 5, DAY 3

Possible Causes of earthquake

- Volcanic activity – magma shifting under the Earth
- Human actions – mining, blasting underground, big constructions
- Tectonic plates moving, mostly where the plates meet

What to do in an earthquake

- Duck cover and hold
- make sure to stay calm
- in streets or cities, do not go outside because of falling debris from buildings
- you can move to very open areas with little to no tall collapsing things

After the earthquake:

- check for injuries and do first aid
- watch out for gas leaks or open live wires

C.3.2.4 | Participant 4 Recollections

TOPIC 1, DAY 1

A cell is a smallest structural and functional unit in life, example of cell a bacterial cell. Some cells are found in the organs or body like blood cell

TOPIC 2, DAY 1

The endocrine system a network that produce hormones, chemicals that message allowing cells and organs to communicate. Hormones can also help control growth, metabolism, reproductive, stress, response, and mood. Gland is also an organ

TOPIC 3, DAY 1

Applied science used for scientific knowledge to solve problems. It also focuses on specific problems, like specific principals and to create practical and useful information

Problem - oriented focus Science

TOPIC 4, DAY 1

Psoriasis a long term skin condition causing itchy,scaly patch that commonly appear on the knees, elbows scalp and trunk it is painful affect sleep and make it hard to concentrate

Types of psoriasis plaque psoriasis cause is dry and scaly patch

Nail psoriasis finger or toenail

TOPIC 5, DAY 1

Earthquake is the shaking of the earths surface. A sudden movement of the earths crust

Cause Tectonic plate movement plate boundaries due to collisions or separation or sliding past on one another

Volcanic activity magma movement beneath the surface

Physical damage building collapsing and people getting injure

TOPIC 1, DAY 2

Characteristics of cells are surrounded by palm membrane, contain cytoplasm, have genetic material like dna rna

Two main types of cells Prokaryotic calls non true nucleus, non membrane bound organelles and eukaryotic cells have many nucleus organs and linear dna

TOPIC 2, DAY 2

Endocrine system has many glands located in the body, such as Hypothalamus are located in the brain and pituitary gland that controls other glands

Thyroid controls growth

TOPIC 3, DAY 2

applied science appears in many fields such as computer science creating software algorithms and friendly apps

Electrical engineering building, electronics and communication system

Mechanical engineering designing engines vehicle robots and machines

Data science can analyze large datas and build ai

TOPIC 4, DAY 2

psoriasis can be developed through family history and smoking can increase risk It can also lead to completions like eye problem obesity high blood pressure hear disease and depression and other illnesses

Trigger includes Infection

TOPIC 5, DAY 2

earthquakes are natural shaking of the earths surface that can lead to a dangerous hazard and can have devastating consequences

During an earthquake we should stop drop and hold on stay calm and and not panic we should

After an earthquake we should check for injuries

TOPIC 1, DAY 3

A cell is the smallest unit of life and has a cytoplasm and a cell membrane. But not all have nucleus and organelles.

Viruses, DNA, and crystals are not cells or are not and out of cells.

Two main types of cells: Prokaryotes are cells with no nucleus and have a circular dna. No membrane organelles Eukaryotes have a nucleus and linear dna. Eukaryotes can do mitosis or meiosis.

Cells move through flagella and cilia which are like tiny hairs.

TOPIC 2, DAY 3

The endocrine system is what produces hormones in the body. It controls growth, or metabolism, or reproduction, and many other.

Different glands:

- Hypothalamus
- Thyroid
- Parathyroid
- Testes
- Ovary
- Pancreas

Risk factors of endocrine imbalance can involve age or genetics.

TOPIC 3, DAY 3

Applied science is using scientific knowledge to solve real problems. It is practical and "problem-focused" approach. It is different from other science because it is applied in real life and uses the scientific method to study existing info and test and improve. And it continues this cycle.

Examples are vaccine, electric cars, and data centers.

Types of applied science:

- Data science chemical science
- Mechanical science
- Electrical engineering
- Computer engineering

TOPIC 4, DAY 3

Psoriasis is a kind of skin disease that appears as scaly and dry patches or cracked skin. Sometimes it burns or itches.

It is caused by either infections, or weather, or unhealthy habits, or even from genetics.

Types of psoriasis:

- Pustular
- Inverse
- Nail
- Plaque

See a doctor when symptoms show

TOPIC 5, DAY 3

Duck cover and hold during earthquake. Prepare emergency kits before earthquakes or at all times in case of earthquakes. Participate in earthquake drills and have evacuation plans. Be careful of falling objects and being near tall structures.

Earthquake is from mining and digging but mainly from the shaking of the earths plates due to tectonic movement.

C.3.2.5 | Participant 5 Recollections

TOPIC 1, DAY 1

A cell is a smallest and a fundamental unit of life that can carry out all the vital biological functions. A cell is not a DNA, mitochondria, crystal, water droplet, and a virus. The cell helps us to live and the cell is not only found in humans but in everything, it can be a plant, animal, bacteria, and a fungi. There are two types of cells: the Prokaryotic Cells and Eukaryotic Cells. Prokaryotic cells are found in bacterias, while the Eukaryotic Cells are found in animals, plants, and fungi. The structure of the cells has (mixed version): cell wall, golgi apparatus, mitochondria.

TOPIC 2, DAY 1

The endocrine gland plays a vital role in our body that helps our body function normally. This helps us balance and maintain our body to have a healthy life. If the endocrine gland is unhealthy it may affect our body, mood, and balance. The parts of the endocrine system are the pineal gland, pituitary gland, thyroid.

TOPIC 3, DAY 1

In today's modern world we often cling to Science for more discoveries, wonders, and awe-inspiring information. Science helps us to produce many inventions and innovations. Applied Sciences appears in many fields like Computer Science, Mechanical Engineering, Biomedical Engineering, Aerospace Engineering, and Biomedical Engineering.

TOPIC 4, DAY 1

Psoriasis is a complex skin infection that may affect the person's well-being. And it may have a cure for it but we need to be careful because prevention is better than cure. And the possible ways to prevent this is to prevent sun exposure, eating healthy, and wearing sunblock, etc. Moreover, there is nothing to worry about because there are professional dermatologists that can help you through it. And pray.

TOPIC 5, DAY 1

Earthquakes are known as a destructive calamity that affects many living things and infrastructures. Earthquakes will happen naturally and unnaturally. And we may record and track earthquakes in a seismograph and it is monitored by the seismologist. We must be prepared and be ready for unexpected earthquakes.

TOPIC 1, DAY 2

The cell is the basic unit of life and it helps all living organisms live and experience life. All living things are made up of cells. Some organisms have only one cell (unicellular), while others have millions (multicellular). Each cell has parts with specific jobs, like the nucleus that controls activities, the cell membrane that protects the cell, and the cytoplasm where many processes happen. Cells work together to keep the body alive and functioning. There are two types of cells we have: the prokaryotic and eukaryotic cells. These cells are not always found in all living organisms, these cells have their own living organisms.

TOPIC 2, DAY 2

The endocrine system is composed of different glands which secretes hormones that regulate metabolism, growth and development, mood and reproduction. If the endocrine system malfunctions our body may have problems in reproduction,

THE ENDOCRINE SYSTEM:

- PINEAL GLAND
- PITUITARY GLAND
- THYROID
- PARATHYROID GLAND
- ADRENAL GLAND
- PANCREAS
- OVARIES
- TESTES

TOPIC 3, DAY 2

Applied Science is the branch of science that uses scientific ideas to solve real-life problems, create technology, and improve everyday life. It is also the usage of scientific knowledge to solve real problems. A key feature of applied Science is Problem oriented-focus. Applied Science appears in many fields: Computer Science, Electric Engineering, Materials Science, Chemical Engineering, Materials Science, Biomedical Engineering, Aerospace Engineering, and Data Science.

TOPIC 4, DAY 2

Psoriasis is a chronic (long-lasting) skin condition where the body's immune system makes skin cells grow too fast. Because of this, thick, red, and scaly patches appear on the skin.

Key points:

It is not contagious — you cannot catch it from someone.

Patches often appear on the elbows, knees, scalp, back, or other parts of the body.

Causes include genetics, immune system problems, and triggers like stress, infection, cold weather, or skin injury.

Treatments help control symptoms, such as creams, light therapy, and oral or injectable medicines.

TOPIC 5, DAY 2

An earthquake is the shaking of the Earth's surface caused by the sudden release of energy in the Earth's crust. This usually happens when tectonic plates move or converge, diverge, and slide past each other creating stress that is released as vibrations called seismic waves.

Earthquake preparedness is the practice of getting ready before, during, and after an earthquake to stay safe and reduce danger. Before an earthquake, people should prepare an emergency kit with water, food, flashlights, medicines, and important documents. They must also secure heavy furniture and identify safe spots like under sturdy tables. During an earthquake, the safest action is to Drop, Cover, and Hold On, staying away from windows and heavy objects. After the shaking stops, it is important to check for injuries, avoid damaged buildings, turn off gas or electricity if needed, and stay alert for aftershocks. Being prepared helps save lives, prevents injuries, and reduces panic during disasters.

TOPIC 1, DAY 3

Cells are the basic building blocks of all living things. Every plant, animal, and human is made up of cells. Some living things have only one cell (like bacteria), while others have millions or even trillions of cells (like humans).

What cells do:

- They give structure to living things.
- They take in nutrients and turn them into energy.
- They help living things grow and repair damaged parts.
- They reproduce to make new cells.

Two main types of cells:

- Animal cells
- Plant cells (these have extra features like a cell wall and chloroplasts)

TOPIC 2, DAY 3

The endocrine system is made up of glands that produce hormones. These hormones travel through the blood and control many body activities such as growth, metabolism, mood, and reproduction. Major glands include the pituitary, thyroid, adrenal glands, pancreas, ovaries, and testes. This system helps keep the body balanced and functioning properly.

TOPIC 3, DAY 3

Applied Science is the use of scientific knowledge to solve real-world problems. It takes ideas from biology, chemistry, physics, and other sciences and applies them to create useful tools, technologies, and solutions. Examples include making medicines, designing machines, improving farming, and developing new materials.

Examples of Applied Science:

1. Medicine – Developing vaccines, antibiotics, and medical equipment.
2. Engineering – Designing bridges, cars, airplanes, and machines.
3. Agriculture – Creating fertilizers, pest control methods, and irrigation systems.
4. Environmental Science – Treating polluted water, recycling, and renewable energy solutions.
5. Technology – Building smartphones, computers, and household appliances.

TOPIC 4, DAY 3

Psoriasis is a chronic (long-term) skin disorder in which the skin cells grow too quickly, forming thick, red, scaly patches. Normally, skin cells grow and shed in about a month, but in psoriasis, this process happens in just a few days. This rapid buildup causes irritation, redness, and sometimes itching or pain. Psoriasis is not contagious, but it is linked to the immune system and genetics. Triggers can include stress, infections, cold weather, or certain medications. Treatment focuses on reducing inflammation, controlling symptoms, and slowing down skin cell growth through creams, ointments, light therapy, or medications.

TOPIC 5, DAY 3

Earthquake preparedness means getting ready in advance to protect yourself, your family, and your property during an earthquake. It involves knowing what to do before, during, and after an earthquake to reduce injuries, save lives, and minimize damage.

Examples of earthquake preparedness:

- Making an emergency plan and practicing drills
- Preparing an emergency kit with food, water, and first-aid supplies
- Securing heavy furniture and appliances to prevent them from falling
- Knowing safe spots in your home (like under a table or against an interior wall)

C.4 | Statistical Analysis Summary

C.4.1 | Comparative Assessment Validation

The dual-method validation between SimCSE and human teacher assessments yielded the following statistical measures:

SimCSE vs. Teacher Assessment: Mean absolute difference: 9.54 points (SD = 7.23) Pearson correlation coefficient: $r = 0.847$ Statistical significance: $p < 0.001$ Cohen's d effect size: 1.24 (large effect)

Assessment Validity: SimCSE demonstrates exceptional alignment with teacher assessments, establishing strong validity for automated semantic similarity evaluation in educational contexts.

C.5 | System Architecture Specifications

C.5.1 | Technical Implementation Details

Frontend Technology Stack: React.js with Next.js framework for server-side rendering optimization, Tailwind CSS for responsive design implementation, Chart.js for real-time data visualization, and RESTful API integration for seamless data communication.

Backend Architecture: Node.js runtime environment with Express.js framework, PostgreSQL database with normalized schema design, SimCSE microservice integration through dedicated API endpoints, and comprehensive session management with secure authentication protocols.

Natural Language Processing Pipeline: BERT-base-uncased model serving as foundational encoder, SimCSE contrastive learning implementation with dropout-based data augmentation, cosine similarity computation for semantic distance measurement, and text preprocessing pipeline for formatting normalization.

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